Smoke Aerosol Measurement Experiment (SAME)

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Abstract

Leaks, fires, and spills are some of the most significant hazards that must be accounted for in the design of the environmental control and life-support system on spacecraft. While prevention is certainly the first line of defense against a fire, early detection of a overheat event or an actual fire allows the crew to respond before the hazard gets out of control. The limited options available for response to a spacecraft fire increase the importance of detecting the fire in an earlier stage of its development than is typical for terrestrial applications. This report describes the outcomes of the Smoke Aerosol Measurements Experiment (SAME), an experiment conducted in the Microgravity Science Glovebox (MSG) on the International Space Station (ISS) to characterize the size distribution of smoke particulate produced in low-gravity from a variety of overheated spacecraft materials The absence of gravity can impact not only the formation of the smoke particulate but also the aging and coagulation processes that occur as the particulate are transported from their source to a detector.

Based on the results of an earlier flight project carried out in the Glovebox of the Space Shuttle, (Comparative Soot Diagnostics (CSD)), the soot particles generated and observed in low gravity environment tend to be larger in size than those in earth conditions. However, the preliminary results of CSD project were not conclusive because of a minor variation of the temperature of hot wire which heated the test sample to generate smoke, and also because the CSD apparatus could not measure the liquid drops existing in the smoke produced by several of the samples. Furthermore, the CSD project was not designed to provide on-orbit measurement of the particle size distribution. The Smoke Aerosol Measurement Experiment was designed to obtain this data that could then be used to design the next-generation smoke detectors for space exploration missions. Unfortunately, because of the relatively small volume of the MSG, it cannot accommodate the instruments normally used in a ground-based laboratory to measure the size distribution of smoke particulate. Therefore, a new procedure was developed, the moment method, from which the characteristics of the particle size distribution can be inferred using a minimum number of portable instrument.

The SAME experiments were conducted during Expedition 15 on the ISS. Four materials, typical of those found in spacecraft, were tested including cellulose (lamp wick), Kapton, silicone rubber, and Teflon. A fifth material, dibutyl-phthalate (DBP) retained in a cotton wick was used as a reference. In order to simulate the detection of a pre-fire overheated-material event, samples were heated to temperatures below the ignition point. The smoke generation conditions were controlled to provide repeatable sample surface temperatures and air flow conditions. The transport time from the smoke source to the detector was simulated by holding the smoke in an aging chamber for times ranging from 10 seconds to 1800 seconds. Smoke particle samples were collected on Transmission Electron Microscope (TEM) grids for post-flight analysis. The smoke properties were measured using particulate aerosol diagnostics that measure different moments of the size distribution. These statistics were combined to determine the geometric mean particle sizes and geometric standard deviation, which is sufficient to describe the entire smoke distribution if a lognormal distribution is assumed. For comparison, smoke detectors from

the ISS and the space shuttle were also included in the SAME apparatus. The experimental results show that, for the materials tested, a substantial portion of the smoke particles are below 500 nm in diameter. The smoke transport model demonstrated that mixing dominates the smoke transport and that consequently detection time lags are longer than in normal gravity. The knowledge gained from SAME can be used to establish design requirements for smoke detectors used in future spacecraft.

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Nomenclature

$\bar{d}$	--	Count mean diameter or arithmetic mean diameter
$d_{\bar{m}}$	--	Diameter of average mass
D	--	Diameter of particles
D_g	--	Geometry mean diameter
D_p	--	A type of characteristic diameter depending on the value of p
$f_N(D)$	--	Frequency of particle count distribution function (zeroth moment distribution function)
$f_i(D)$	--	Frequency of the *i-th* moment distribution function
$F_i(D)$	--	Cumulative *i-th* moment distribution function
L	--	Letter for length dimension
M_i	--	The *i-th* moment
$N(D)$	--	Number of particles per unit volume with diameters equal or less than D
N_t	--	Total number of particles per unit volume
$u(x)$	--	Absolute uncertainty of variable x, with the same unit as variable x
$u_r(x)$	--	Relative uncertainty of variable x, dimensionless. $u_r(x) = u(x)/x$
σ_g	--	Geometric standard deviation of a lognormal distribution

Acronyms

CMD: Count Mean Diameter
CSD: Comparative Soot Diagnostics Experiment
DACU: Data Acquisition and Control Unit
DAFT: Dust Aerosol measurement Feasibility Test, a precursor project of SAME
GMD: Geometric Mean Diameter
GSD: Geometric Standard Deviation
ISS: International Space Station
MSG: Microgravity Science Glovebox
SAME: Smoke Aerosol Measurement Experiment
STS: Space Transportation System
TEM: Transmission Electron Microscope

1. Introduction

Appropriate design of fire detection systems requires knowledge of both the expected signature of the events to be detected and the background levels of the measured parameters. Terrestrial fire detection systems have been developed based on extensive study of terrestrial fires.[1, 2] Unfortunately, there is no corresponding data set for spacecraft fires and consequently current spacecraft fire detectors were developed based upon experience on normal gravity fire behavior and properties. This approach was necessary given the absence of data concerning microgravity smoke particulate. However planned spacecraft and their planned missions outside of low earth-orbit will require increased reliability of fire detection systems in addition to robust false alarm resistance. Extended lunar-surface missions will require systems that are not influenced by lunar dust and other nuisance dust sources.

The absence of buoyant flow increases the residence time in microgravity fires causing substantial changes in flame structure. Given the large changes in fire behavior seen in reduced gravity, it is reasonable to expect that the signatures and detection requirements of these fires will change. There are a number of factors that can be expected to affect the particle size distribution of the smoke from spacecraft fires. The absence of buoyant flow in low-gravity increases the residence time in microgravity fires and increases the transit time from the reaction zone to the detector.[3] The materials used in spacecraft are different from those typically used in terrestrial applications where smoke properties were previously evaluated. More importantly, the limited options available to respond to a spacecraft fire increase the importance of detecting the fire in an earlier stage of its development than is typical for terrestrial applications. All of these effects can be expected to change the smoke particle size distribution. The objective of this work was to make sufficient measurements of smoke from spacecraft fires to enable improved design of future detectors.

2. Smoke Background

Smoke is a general term that encompasses aerosol materials produced by a number of processes. In particular for non-flaming combustion, the aerosol includes recondensed original material or pyrolysis products that can be either liquid or solid. For flaming combustion, the aerosol includes soot particulate, condensed water vapor, and ash particles. Soot particles dominate the smoke particulate in established flaming fires while unburned pyrolysis products and recondensed polymer fragments are produced by smoldering and pyrolysis in the early stage of fire growth. Given the constrained space on any spacecraft, the target for the fire detection system is necessarily the early phase and not established flaming fires; consequently, the primary target for detection is the pyrolysis products and not the soot.

The background regarding fire detection in US spacecraft is summarized in this paper but is discussed in more detail in papers by Friedman and Urban[4, 5] In the Mercury, Gemini and Apollo missions, the crew quarters were limited and mission durations were short, consequently it was considered reasonable that the astronauts would rapidly detect any fire. The Skylab module, however, included approximately 30 UV-sensing fire detectors.[4] These devices were limited to line-of-sight and were reported to have difficulties with false alarms.

Details of the ISS and STS detector requirements are described by Steisslinger et al.[6] A brief overview of the requirements is given below.

Space Shuttle Detectors:

At the time the Space Shuttle Orbiter (Space Transportation System (STS)) was developed, smoke detection for terrestrial applications was becoming commonplace.[1] In particular, ionization detectors were readily available but photoelectric (scattering or obscuration detectors) were generally unavailable due to the difficulty of producing stable light sources. There were no data available concerning the smoke or dust particle size distribution in low-gravity and the database for normal-gravity smoke characteristics was a fraction of what is available today. Consequently, the STS smoke detector (Fig. 1) employed a variant of the accepted ground-based approach (ionization detectors). The Orbiter has nine particle-ionization smoke detectors in avionics cooling air return lines in the mid-deck and flight deck and the SpaceLab had six additional particle-ionization smoke detectors in the avionics lines.[7] While the design rationale is not completely known, Celesco™ (later Brunswick Defense™) based their design on data that suggested incipient fires could be discriminated by looking for particulate in the 0.4 μm to 0.7 μm range.[8] The use of a pump provided the opportunity to use a particle separator that rejected particulate larger than 1μm. This air pump increased the power requirements and reduced the operational life of the detector. The STS detector as built was designed to alarm at a particle loading of 2 mg/m^3 for 5 seconds (based on 1 micron particles) or a 0.022 mg/m^3/s rise in concentration for 20 seconds[9].

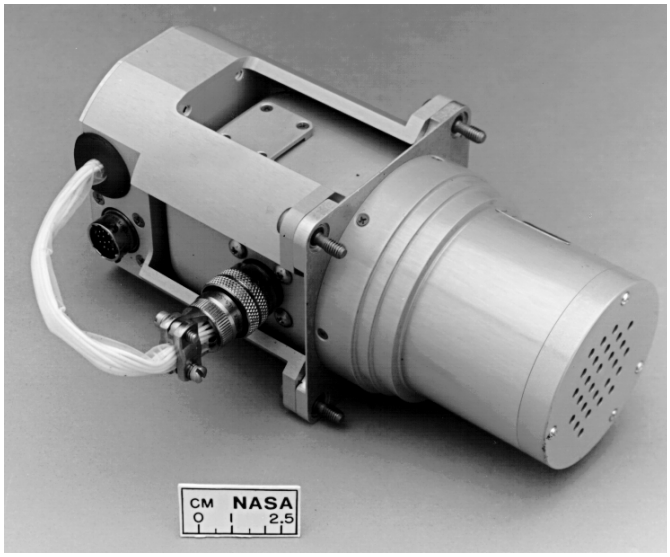


Figure 1. Brunswick Defense™ smoke detector used in the NASA orbiter fleet. The inlet is on the right and the gas is expelled out the small plate on the top.

ISS Detectors:

The International Space Station (ISS) smoke detector design took advantage of the availability of stable laser-diode light sources, mirroring a trend in terrestrial detectors toward light scattering devices. This provided the opportunity to produce a detector that required substantially less power, (1.5 Watts versus 9 Watts for the orbiter detector)[6] and, with no moving parts, was quieter with a much longer operational life. The choice was supported by data indicating that early smoldering fires produce larger particulate than established flaming fires.[1] The ISS detectors (Fig. 2) are laser-diode, forward-scattering, smoke or particulate detectors. The sensitivity extends down to 0.3 μm but is greatest for particles larger than 1 μm. The current requirements for the ISS call for two detectors in the open area of the module, and detectors in racks that have cooling air-flow.[10]

The design developed for the ISS by Allied Signal™ (Fig. 2) consists of a 2-pass laser diode to sense forward scattered light (30 degrees). There is also a light transmission sensor to monitor the beam intensity. The system is designed to alarm based on the magnitude of the scattered light signal referenced to the beam intensity. The system is less sensitive to particles smaller than the wavelength of the laser (near IR) than it is to larger particles. The minimum reported sensitivity is 0.3 μm.[6] The ISS detector was designed to alarm at 1% obscuration per foot using an Underwriters Laboratory (UL)

smoke box and white light extinction. This was implemented using a transfer standard detector and a 0.5 μm polystyrene-bead aerosol system that was used to set the amplifiers on each unit. The transfer standard was calibrated in the smoke box and then used to set the levels with the aerosol system.

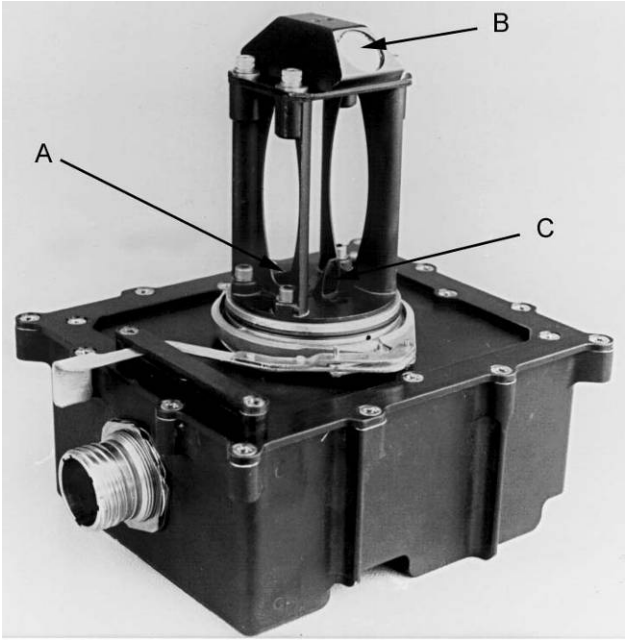


Figure 2. Allied Signal/Honeywell light-scattering smoke detector used in the ISS. The near IR laser beam emerges from the enclosure into the top assembly (A) and is reflected by two mirrors (one visible at top right (B)) and is then reflected back to the sensors in the enclosure (C). One sensor detects the forward scattered light and is referenced by another sensor that looks directly at the incident beam.

As a result of these design decisions, the ISS and the Orbiter detectors have particle sensitivities that do not overlap. In both cases the designs were based upon the best available data at the time. However, due to the complete absence of low-g data concerning the nature of particulate and radiant emission from incipient and fully developed low-g fires, different conclusions were drawn concerning the optimal design for spacecraft fire detection.

As described by Friedman[4] there have been six overheat and failed component failures in the NASA Orbiter fleet in addition to several similar incidents that have occurred on the ISS. None of these events spread into a real fire but as mission durations increase, the likelihood of failures increases. The experience on Mir in 1997 has shown that failure of oxygen generation systems can have significant consequences. As a result, improved understanding of spacecraft fire detection is critically needed.[11]

Previous work on smoke particles from low-gravity sources by Urban et al.[5] indicated that the particulate produced by low-gravity flames (soot or unburned fuel particles) tends to have larger size particles than in normal gravity. Results from the CSD (Comparative Soot Diagnostics) Experiment[5] which studied smoke properties in low-gravity from several spacecraft materials suggested that liquid smoke particles could achieve sizes larger than 1 μm while solid particulate remained in the sub-micrometer range. However, the CSD experiment did not produce sufficient data concerning the size of the liquid smoke particles to guide detector design. The combined impact of these limited results and physical expectations is that, based on the increased smoke residence time in low gravity, direct knowledge of low-g combustion particulate is needed for the confident design of smoke detectors for spacecraft rather than extrapolation from 1-g data

Any effort to detect fires must be able to discriminate against the ambient background and nuisance signals. In the case of smoke detection, the background dust aerosol conditions must be considered. The only spacecraft background aerosol particulate measurements to date were made on the Orbiter on STS-32 in 1990[12] and on the ISS.[5]

The STS measurements included the results from two cascade impactors and a light scattering device. The two impactors reported a bimodal particle size distribution with ~ 40% of the particles in each of the 2.5 μm to 10 μm and >100 μm ranges. The other two ranges <2.5 μm and 10 μm to 100 μm were very lightly populated. Each impactor sampled approximately 15 m^3 of air over approximately 30 hours. These results showed substantially higher concentrations than typical indoor measurements (over a factor of 5 for the two most populated size ranges). These data were supported by the light scattering instrument which made 17 measurements in 12 locations, all of these reported mass concentrations ranging from 50 $\mu g/m^3$ to 70 $\mu g/m^3$. The ISS results of Urban et al.[5] were obtained from a feasibility experiment to validate the operation of an aerosol particle counter in low-gravity (DAFT). The measurements were made in the US Lab on Increment 10 and the particle count levels were at the noise level for the instrument (less than 5 particles/cm^3). The low particle levels in the < 2.5 μm bin for the STS and the low overall numbers for the ISS suggest a zone of opportunity for spacecraft fire detection since typical normal gravity fires produce substantial particulate in this size range[5,2] However more complete particle size statistics are needed.

3. SAME Experiment

To address the limited data from the CSD experiment concerning the likely size of spacecraft smoke particulate, the Smoke Aerosol Measurement Experiment (SAME) was developed. The SAME experiment sought to avoid the problems experienced by the CSD experiment by obtaining the particulate size statistics on-orbit with a reduced dependence upon sample return to Earth. This is a challenging endeavor because existing aerosol instrumentation is typically too large to be compatible with spacecraft experiment constraints. Thus, to obtain particulate size statistics with existing aerosol instrumentation would require the return of numerous samples to Earth. As will be described below, an alternative approach was employed that used three discrete instruments to measure separate moments of the size distribution. When combined, these moments provide useful aggregate statistics of the size distribution if a lognormal distribution is assumed. The measurements were made using smoke generated by overheated spacecraft materials in much the same manner as the CSD experiment however the sample temperature, flow field, and particle aging time were more rigorously controlled.

A. The Moment Method

The approach used by the SAME experiment is termed the 'moment method' for convenience.[13] This approach consists of measuring three moments of the size distribution (the zeroth, first and third moment) on orbit, using several small, light-weight instruments. By applying the properties of the lognormal distribution to the measured moments, the total number concentration, N_t, the geometric mean diameter, D_g, and the geometric standard deviation, σ_g, of the aerosol can be estimated. Once these three parameters are known, the size distribution of the measured aerosol is uniquely defined, assuming that the distribution is lognormal. In what follows, we will explain how the moment method works.

A.1. Principle of the moment method

The number (or count) distribution of an aerosol describes the number of particles that fall into a certain size range in a unit volume of aerosol. The number distribution is sometimes also called size distribution. Mathematically, a distribution function can be expressed in four different, but related, forms. First, a distribution function can be expressed as a frequency (or density) distribution function or a cumulative distribution function. Then, each of the two can be expressed in terms of its normalized form or non-

normalized form. The normalized forms of any distribution function are much like the probability functions in statistics. To avoid future confusion, we will define these distribution functions first.

The particle number distribution function of an aerosol, $f_N(D)$, is defined as follows:

(1) $$f_N(D) = \frac{dN}{dD} \quad ,$$

where dN is the number of particles per unit volume having diameters between D and $D + dD$. Here, the subscript N in $f_N(D)$ denotes particle numbers. The function $f_N(D)$ is a non-normalized frequency distribution function with a dimension of $[L^{-4}]$. The corresponding cumulative distribution function, $F(D)$, is defined as a definite integral of $f_N(D)$ as follows:

(1-a) $$F(D) = \int_0^D f_N(\xi) d\xi$$

As the upper limit D of the integral in Eq. (1-a) approaches infinity, $F(D)$ approaches the total number of particles in a unit volume, denoted by N_t. Thus, $F(D)$ is a non-normalized cumulative distribution function, with a dimension of $[L^{-3}]$. When $f_N(D)$ or $F(D)$ is normalized by N_t, one gets its normalized counterpart. Note that N_t has a dimension of $[L^{-3}]$, thus the normalized cumulative distribution function is always dimensionless whereas the normalized frequency distribution function has a dimension of $[L^{-1}]$. In this report, unless otherwise stated, we will use non-normalized distribution functions, either in their frequency forms or in their cumulative forms.

Based on $f_N(D)$, distribution functions of various properties of a given aerosol can be defined. For instance, the distribution functions of particle diameter, $f_D(D)$, surface area, $f_A(D)$, and particle volume, $f_V(D)$ (or mass distribution function, $f_m(D)$ if the particle material density is constant) are defined in the following manner.

(2) $$f_D(D) = f_1(D) = D\frac{dN}{dD} = Df_N(D)$$

(2a) $$f_A(D) = \pi f_2(D) = \pi D^2 \frac{dN}{dD} = \pi D^2 f_N(D)$$

(2b) $$f_V(D) = \frac{\pi}{6} f_3(D) = \frac{\pi}{6} D^3 \frac{dN}{dD} = \frac{\pi}{6} D^3 f_N(D)$$

(2c) $$f_m(D) = \frac{\pi}{6} \rho f_3(D) = \frac{\pi}{6} \rho D^3 \frac{dN}{dD} = \frac{\pi}{6} \rho D^3 f_N(D)$$

It can be seen that to address various aerosol property distributions, it is convenient to define a moment distribution function $f_i(D)$ as follows.

(2d) $$f_i(D) = D^i f_N(D) \qquad i = 0, 1, 2, ...$$

When $i = 0$, $f_0(D)$ and the number (or count) distribution function, $f_N(D)$ are identical. Because equation (2d) is similar to the calculation of moment in mechanics, $f_i(D)$ is conventionally called the *i-th* moment

distribution function of the aerosol. Note that each $f_i(D)$ is a frequency distribution function. The corresponding cumulative distribution function, $F_i(D)$, is as follows.

(2e) $$F_i(D) = \int_0^D \xi^i f_N(\xi) d\xi \qquad i = 0, 1, 2, ...$$

When the upper limit of the above integral approaches infinity, the integral of (2e) returns a single value, which is called the *i-th* moment of the distribution function $f_N(D)$, denoted by M_i. That is

(2f) $$M_i = F_i(\infty) = \int_0^\infty \xi^i f_N(\xi) d\xi, \qquad i = 0, 1, 2, ...$$

The dimensions of $f_i(D)$ are $[L^{i-4}]$ and those of $F_i(D)$ and M_i are $[L^{i-3}]$. Note that for the sake of generalization, we do not set a limit on the number of moments now, although the highest moment used in SAME is the third moment.

If $f_N(D)$ is an arbitrary function, all its moments are independent of each other making it impossible to reconstruct $f_N(D)$ from a finite number of its moments. However, if $f_N(D)$ is a function that can be uniquely defined by a finite number of parameters, only that same number of its moments are independent and they can be arbitrarily selected. From any set of independent moments, one can solve for the parameters that define the distribution function, $f_N(D)$. Thus, this type of distribution function, $f_N(D)$, can always be constructed if the independent moments are known. This is the underlying idea of the moment method.

The lognormal distribution function below is widely accepted for describing the size distribution of aerosols[14, 15].

(3) $$f_N(D) = \frac{N_t}{\sqrt{2\pi} D \ln \sigma_g} \exp\left(- \frac{\left(\ln D - \ln D_g\right)^2}{2 \ln^2 \sigma_g} \right),$$

where N_t is the total number of particles in a unit volume, or total number concentration of the aerosol, and parameters D_g and σ_g are the geometric mean diameter and geometric standard deviation, respectively, defined by

(4) $$\ln D_g = \frac{1}{N_t} \int_0^\infty (\ln D) f_n(D) dD$$

and

(5) $$\ln \sigma_g = \left[\frac{1}{N_t - 1} \int_0^\infty \left(\ln D - \ln D_g\right)^2 f_N(D) dD \right]^{0.5}$$

The lognormal distribution is related to the well known normal (or Gaussian) distribution function in that *lnD* is used as the independent variable in the Gaussian distribution, including *dN/d(lnD)* that defines $f_N(D)$, because the particle diameter D can never take negative values.

The SAME project assumes a lognormal distribution function for the aerosol size distribution. Since a lognormal distribution function is uniquely defined by three parameters, N_t, D_g and σ_g, only three moments are needed to reconstruct the distribution function. In SAME, these three moments are chosen to be the zeroth moment, the first moment and the third moment. Based on Eq. (2f), the zeroth moment, M_0 is simply equal to N_t, whereas the first moment, M_1, can be thought of as the "diameter concentration" or integrated diameter per unit volume. The third moment, M_3, is proportional, but not equal, to the total volume or mass of the particles per unit volume of the aerosol. From these three moments, two characteristic diameters are defined. They are a) the count (arithmetic) mean diameter, which is equal to M_1/M_0 and denoted by $D_{0.5}$ or $\bar{d}$ and b) the diameter of average mass[16], which is equal to $(M_3/M_0)^{1/3}$ and denoted by $D_{1.5}$ or $d_{\bar{m}}$. The subscripts of 0.5 and 1.5 used for these two diameters follow an existing naming convention.

There is a unique feature of the lognormal distribution, which is described by the Hatch-Choate conversion equations[17]. The diameters for various moment distributions are related to the geometric mean diameter, D_g, and the geometric standard deviation, σ_g, via the following equation.[14, 15]

$$D_p = D_g \exp\left(p \ln^2 \sigma_g\right) \tag{6}$$

In equation (6), p is a parameter defining the meaning of diameter D_p. For instance, when $p = 0.5$, D_p is the count mean diameter, whereas when $p = 1.5$, D_p is the diameter of average mass. This feature of the lognormal distribution greatly simplifies the application of the moment method in finding the aerosol size distribution. Figure 3 shows a typical lognormal distribution along with various characteristic diameters, where D_g=*1.0* μm and σ_g= *1.6*. Differing from Eq. (3), the y-axis in Fig. 3 is normalized by N_t, i.e., $f_N(D)/N_t$. In this distribution, the corresponding values of the count mean diameter, $D_{0.5}$, and the diameter of average mass, $D_{1.5}$, can be computed from equation (6). They are 1.117 μm and 1.393 μm, respectively.

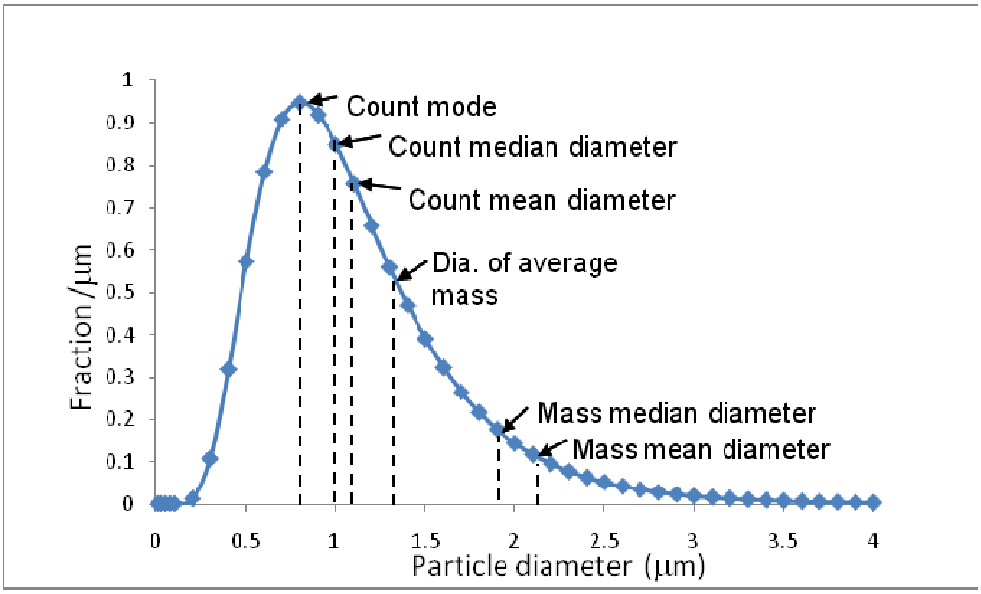


Figure 3: Diameters for a log-normal distribution with D_g=1.0 μm and σ_g= 1.6.

In the case of the SAME experiment, our objective is to obtain D_g, σ_g, and N_t based on measured moments, M_0, M_1 and M_3 with $D_{0.5}$ and $D_{1.5}$ being intermediate parameters. For this purpose, equation (6) is rearranged to provide the following explicit expressions for D_g and σ_g.

(7) $$D_g = \sqrt{\frac{D_{0.5}^3}{D_{1.5}}}$$

and

(8) $$\sigma_g = \exp\left(\sqrt{\ln D_{1.5} - \ln D_{0.5}}\right) = \exp\left(\sqrt{\ln\left(D_{1.5} / D_{0.5}\right)}\right)$$

Thus, once the three moments, M_0, M_1 and M_3, are obtained, we can compute $D_{0.5}$ and $D_{1.5}$. Using equations (7) and (8), we can get the corresponding values for D_g and σ_g. When D_g, σ_g and N_t (i.e., M_0) are known, a lognormal distribution function, $f_N(D)$ can be uniquely defined for the measured aerosol. Validation of this approach is discussed in Cleary, Weinert and Mulholland.[13] This moment method provides a strong basis for the design of the SAME experiment.

A.2. Implementation of the moment method in SAME experiment

In the SAME experiment, the zeroth moment, M_0, was measured using a condensation nuclei counter called the P-Trak™ (TSI Inc.). The first moment, M_1, was approximately proportional to the output signal of an ionization detector, which was a modified ionization chamber taken from a commercially-available smoke detector. The third moment, M_3, was inferred from the light scattered by the smoke aerosol when illuminated by a red laser. For particles in the size range of 0.2 μm to 2.0 μm, , the light scattering signal at 90 degrees from a red laser is approximately proportional to the third moment. The instrument used to obtain the third moment is called a DustTrak™ (TSI Inc.). Calibration measurements were made for the DustTrak to account for the departure of the scattering signal from the 3rd moment. More detailed information of instruments and their operations can be found in the Instruments section below. The instruments used in SAME were compact enough to be installed in the ISS MSG.

Extensive calibrations of these instruments were performed at NIST to ensure that correct values of each moment were obtained from the instrument output signals. The results of these calibrations showed that the instrument signals correlate well with corresponding moments except for the third moment. The third moment was found to be a function of the DustTrak output and the particle diameters. This led to an iterative procedure to find the best-fit results of D_g and σ_g.

A.3. Uncertainty Analysis

Inevitably there are uncertainties in the measured moments. It is interesting to know the contribution of the uncertainty of each moment to the uncertainties of D_g and σ_g. To do so, we need to express D_g and σ_g in terms of the three moments. By substituting $D_{0.5}$ and $D_{1.5}$ by M_0, M_1 and M_3 in equations (7) and (8), we have

(9) $$D_g = M_0^{-4/3} M_1^{1.5} M_3^{-1/6}$$

and

(10) $$\sigma_g = \exp\left(\sqrt{\ln\left(\frac{M_0^{2/3} M_3^{1/3}}{M_1}\right)}\right)$$

The law of propagation of uncertainty[18] for the case of independent variables (i.e., there is no covariance) states

(11) $$u_c^2(y) = \sum_{i=1}^{n}\left[\frac{\partial f}{\partial x_i}\right]^2 u^2(x_i),$$

where $u(x_i)$ is the uncertainty of independent variable x_i, n is the number of independent variables, and $u_c(y)$ is the combined uncertainty of dependent variable $y = f(x_i)$. Uncertainties can be expressed in terms of an absolute value of the variable or, in a relative sense, the ratio of the uncertainty to the value of the variable. Since relative uncertainties are easier to use, we will discuss relative uncertainties in the following text. A subscript r will be added to the letter u to indicate relative uncertainties. Then Eq.(11) can be rewritten as

(11a) $$u_{rc}^2(y) = \sum_{i=1}^{n}\left[\frac{x_i}{y}\frac{\partial f}{\partial x_i}\right]^2 u_r^2(x_i)$$

In our case, y is either D_g or σ_g and x_i $(i = 1, 2, 3)$ are M_0, M_1 and M_3, respectively. Substituting Eqs. (9) and (10) into Eq. (11a), we obtain the relative combined uncertainties of D_g and σ_g as follows.

(12) $$u_{rc}^2(D_g) = \frac{16}{9}u_r^2(M_0) + 2.25u_r^2(M_1) + \frac{1}{36}u_r^2(M_3)$$

and

(13) $$u_{rc}^2(\sigma_g) = \frac{1}{9(\ln\sigma_g)^2}\left[u_r^2(M_0) + 2.25u_r^2(M_1) + 0.25u_r^2(M_3)\right]$$

Equations (12) and (13) show the relative uncertainties of D_g and σ_g as functions relative uncertainties of M_0, M_1 and M_3. It is interesting that both equations (12) and (13) show the uncertainty of M_3 contributes the least to the combined uncertainties of D_g and σ_g, whereas that of M_1 contributes the most.

B. Instrumentation

The moment measurements were made using an assembly of three separate instruments. Two are industrial hygiene instruments manufactured by TSI and one is a modified residential smoke detector.

The zeroth-moment instrument is a condensation nuclei counter, the P-Trak™ (TSI Inc.). This device operates by passing the aerosol-laden air stream through a region saturated with isopropanol vapor and then into a cooler region where the vapor condenses onto the particles increasing their diameter such that they can be readily counted by a light scattering device. This instrument is very robust and operates over a range of 0 to 10^5 particles/cm^3 and 20 nm to 1 μm diameter. Some dilution is required since the smoke concentration in SAME ranges from about 0.5×10^6 to 5×10^6 particles/cm^3. There was also a concern that the isopropanol condensate would not return to the wick in low-gravity.[5] To mitigate this

issue, the condensing section of the device was modified with very small grooves to improve conductance of the condensate back to the wick. These changes were tested in a separate space experiment with good results indicating the modified device could be used successfully in low gravity.[5]

The first-moment instrument is the ionization chamber taken from a residential smoke detector. This device uses an alpha-particle emitter to generate ions in a region within a DC electric field. Normally, the ions drift toward an electrode inside the ionization chamber producing a current. When particulate are present, the ions attach themselves to the particles thereby charging the particles. Because the charged particles have a much smaller mobility than ions alone, the current is reduced. The required particle concentrations are on the order of 10^5 particles/cm^3, thus no sample dilution was required.

The third-moment instrument is a light scattering device, the DustTrak™ (TSI Inc.). The device uses a 90 degree light scattering signal to quantify the aerosol mass density. For terrestrial dust particulate, this signal correlates well with the mass concentration, however additional compensation was needed to account for the range of particle sizes observed in the SAME experiment. The device operating range is from 0.001 mg/m^3 to 100 mg /m^3. These devices are equipped with an aerodynamic impactor at the inlet which captures particles larger than the selected size. The SAME experiment included 2 DustTraks™, one with a 1 μm impactor and the other with a 10 μm impactor. The difference in the signal from the two devices provided a measure of the fraction of the particulate that was larger than 1 μm. In some cases dilution was required owing to the high smoke concentration levels.

A schematic of the assembled hardware appears in Fig. 4. The system was installed in the Microgravity Science Glovebox, an ISS facility that provides many resources including: containment, power, data, video and uplink commanding. Smoke was generated by resistively heating small samples of spacecraft materials. Sample heating was achieved by heating a steel wire that was wrapped around an approximately 3 mm diameter cylinder of the sample material. The steel wire was resistively heated and

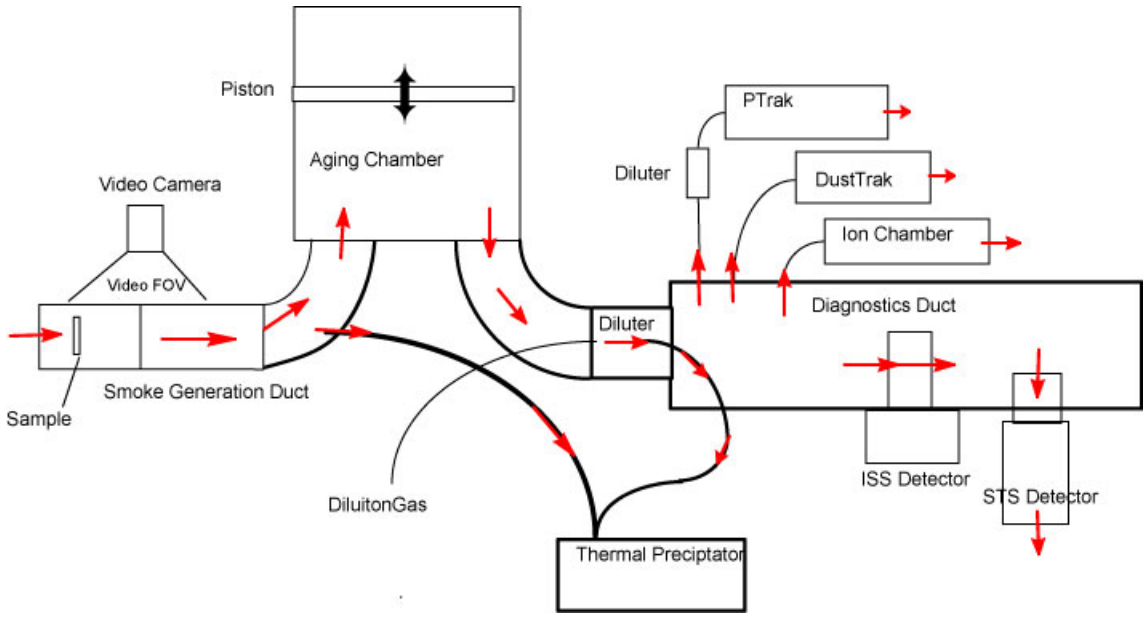


Figure 4. Schematic of the SAME hardware**.**

the wire resistance was monitored to control its temperature. This technique was found to produce repeatable weight loss from sample to sample in laboratory tests. Groups of six pre-weighed samples were installed in carousels (Fig. 5) that the crew installed in the SAME hardware. For each test, a new sample was moved into place by rotating one of the six sample chambers in the carousel into the flow duct and the sample heating and the flow were synchronized to provide a steady air stream over the sample during the heating interval. The system was designed to provide air flow rates ranging from 2 to 8 cm/s.

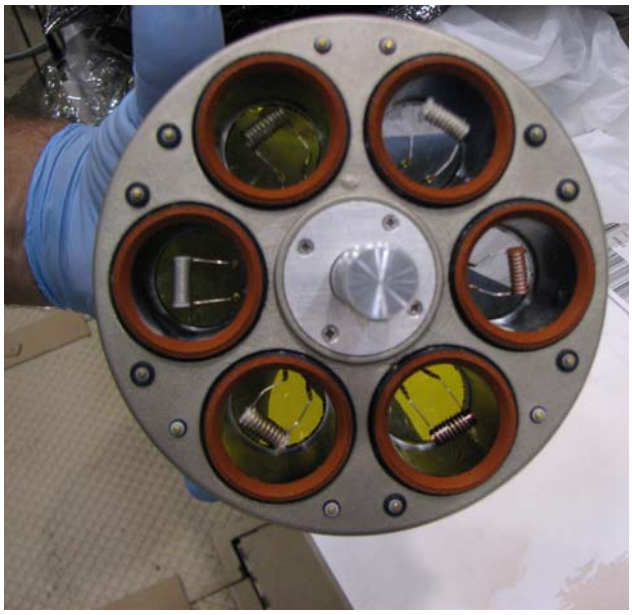


Figure 5. Photograph of a sample carousel. Carousel contains an assortment of Teflon, Kapton, lamp wick, and silicone rubber. The samples were supported and heated by the same stainless steel wire coil.

To emulate a variety of fire scenarios, tests were conducted at different sample temperatures, air flow rates, and smoke aging durations. The sample heating time is approximately 60 seconds. During this interval, controlled flow was induced by a moving piston in the aging chamber which drew the smoke into the chamber. The smoke was held in the chamber for a pre-determined time, allowing the particles to coagulate. After a specified aging time, the smoke was then pushed by the piston into the diagnostics duct where the moment instruments made their measurements. Also installed in the diagnostics duct were space shuttle and ISS smoke detectors. Their output signals were used to compare with the results of the moment method.

As the aging chamber was filled and as the aerosol was pushed out to the moment instruments, a small aerosol stream was drawn off to the thermal precipitator, (Fig. 6). Inside the precipitator, the aerosol passes between a Transmission Electron Microscope (TEM) grid and a heated wire. Thermophoretic

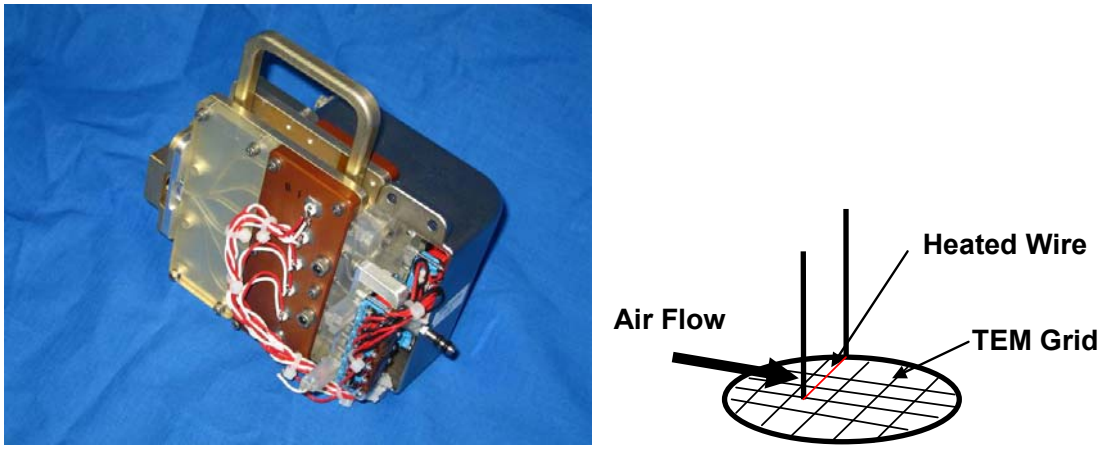


Figure 6. Thermal Precipitator Module. Left image is a photograph of a partially disassembled Thermal Precipitator Module with the inlet manifold exposed on the left, the hot wire leads in the red Vespel block in the middle and the outlet valves on the right. The drawing on the right displays the position of the hot wire above the TEM grid with the air flow passing over both. The heated wire drives aerosol particles onto the grid via thermophoresis.

forces moved particles to the grid surface where they adhered. After the mission the grids were removed from the assembly and examined in a TEM to obtain an independent determination of the particle size and morphology.

C. Instrument Calibration

Calibration of these instruments was essential to properly interpret the flight data and was performed on the ground before the flight. Calibration was accomplished using two different aerosol generators; one generates mono-disperse particle using dioctyl phthalate (DOP) according to the approach by Mulholland and Liu[19] and the other uses polystyrene spheres. The monodisperse droplet generator functioned by producing a spray of DOP diluted with isopropanol which is then evaporated and recondensed producing monodisperse droplets. The droplet size is controlled by the DOP dilution level. The generator will operate stably for tens of minutes. The aerosol from the generator was sampled simultaneously by the SAME instrument under test and a reference instrument. For the number count, the reference instrument was a TSI 3022 particle counter, for the mass concentration of the aerosol, a Tapered Element Oscillating Microbalance and for the first moment a TSI Electrical Aerosol Detector was used. The results for the P-Trak are shown in Fig. 7. As the number concentration increases, the effect of the particle diameter becomes more evident. Separate correlations were developed for each particle size and the closest correlation was used to analyze the flight data based on the initial estimates of the average particle size.

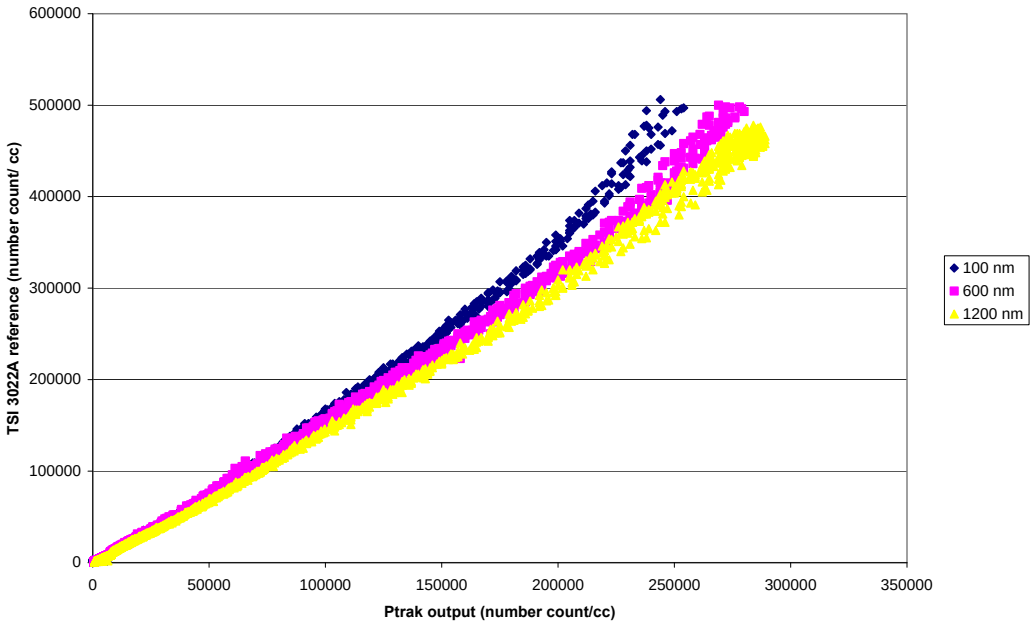


Figure 7. P-trak Calibration results with monodisperse DOP droplets.

The first moment device, the ion chamber, showed very little effect of particle size, as seen in Fig 8. Consequently a single correlation was used for all particle sizes. The third moment device (TSI-DustTrak®) is theoretically predicted to show non-monotonic behavior as particle size is increased.

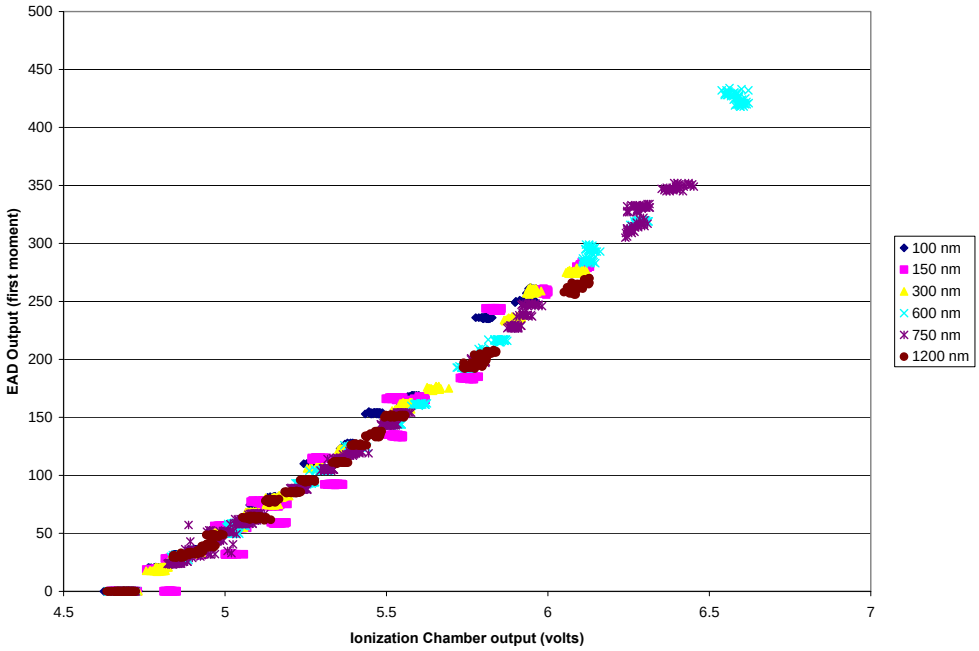


Figure 8. Ion Chamber Calibration results with monodisperse DOP droplets.

While a satisfactory prediction of the DustTrak® response was obtained using liquid droplets of dioctyl phthalate, (Fig. 9) the uncertainty associated with the Mie scattering calculations due to the particle refractive index estimate, the potential of agglomerated particles, and deviation from assumed log-

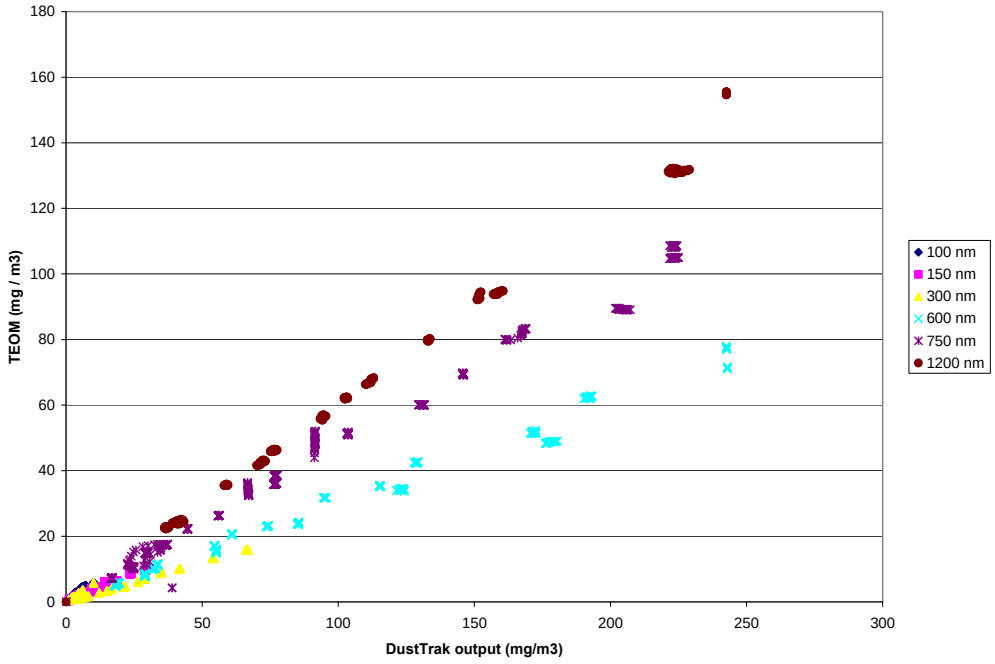


Figure 9. DustTrak results with monodisperse DOP droplets.

normal size distribution must be assessed further for the smoke samples here. In the results reported here, the DustTrak® response was directly calibrated, for each smoke source, against mass concentration measurements using a tapered element oscillating microbalance. This approach is valid as long as the size distribution of the sample does not change significantly. Using estimated material densities, the diameter of average mass was computed for each sample. Equation 8 was used to compute the geometric standard deviation from the arithmetic mean diameter and the diameter of average mass, and Equation 7 was used to compute the GMD.

4. Flight Tests

The SAME hardware installed in the ISS MSG is shown in Fig 10. The SAME tests were performed on board ISS in September and early October of 2007. Overall 30 sample materials were tested. These were comprised of six samples each of 5 materials: Teflon™, Kapton™, silicone rubber, cellulose, and dibutyl-phthalate deposited on a porous wick. These samples were tested at different airflow rates, pyrolysis temperatures, and smoke aging durations.

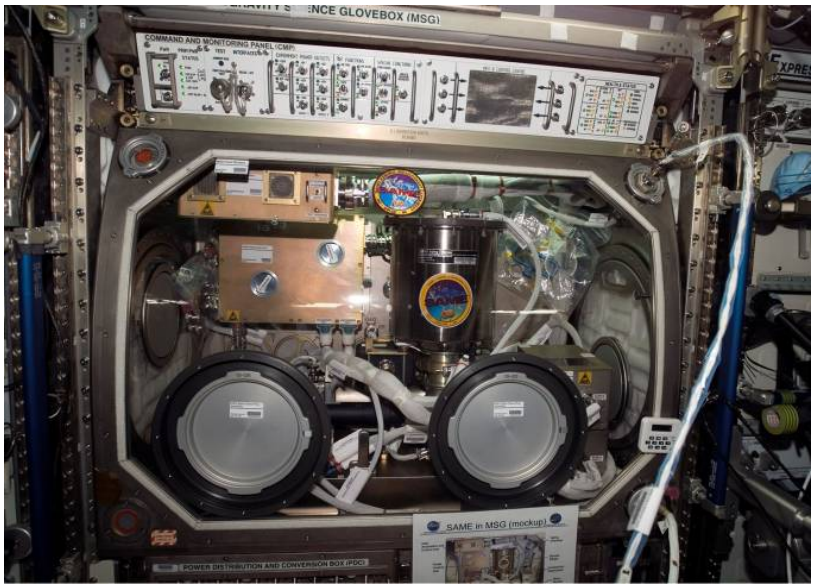


Figure 10. SAME Experiment installed in the ISS MSG Facility**.**

Realizing that crew time was the most precious ISS resource, SAME was designed to be operated both autonomously and from the ground to the greatest extent possible. The SAME hardware was organized into a number of smaller modular units to facilitate stowage for launch, provide for easy change out, increase the reliability and reduce risk. The downside of this strategy was that the assembly of the experiment involved many connections and required approximately 5 hours of crew time. Once installed, however, SAME was easily and effectively operated from the Tele Science Center (TSC) at the Glenn Research Center (GRC), and this capability enabled the ground team to achieve roughly a 50% greater number of test points (and data) than was originally planned.

The on orbit operations were scheduled around the availability of the crew for installing the next Sample Carousel and Thermal Precipitator (and removing and stowing the used ones). Typically, the crew would install the next units during the morning, and operations would be scheduled for later that day. SAME would be nominally scheduled by the ISS staff for two test points on each of the following days.

Because the allowed operations windows were sufficiently large, the ground team was frequently able to perform four test points in the allowed time.

The staffing level of the ground team varied between two and five members from the engineering team with additional support from up to three members of the Science team. Most of the time, there would be one person operating the linked computer that controlled the experiment, one providing verbal communication to the ISS cadre and doing data reduction and a third recording video and performing other miscellaneous functions.

Uplink parameter sheets were generated by the Science team and most times, emailed to the engineering team the day before operations, although when additional unplanned test points were performed during a session, the Science team provided a hard copy of the parameters during the interim between test points. These sheets were printed and checked, and in most cases, signed. The parameters were entered into the ground computer by a team member and then the entry was checked both by means of a checksum and by the other member on consol before uplinking them to the DACU.

The test points were scheduled by the ground team to coincide with Ku and S band coverage so that video of the heating and the measurement data could be obtained Near Real Time (NRT) and recorded on the ground. Because the data reduction software was already in place, processing of the data could be done immediately after the test point had been completed, and in all cases, before the next test point was performed. Periodically during the overall operations, downloads from the DACU were commanded as a backup to the NRT data to insure that the most complete compilation was available.

Detailed chronological record of the SAME flight tests can be found in Appendix A.

5. Test Results and Discussions

The analog signals from various instruments were digitized at 1 Hz by an 8 bit A/D converter. Figure 11 contains the signal traces from a typical run for silicone rubber. The difference in the signals for the two DustTrak's indicates that a substantial portion of the aerosol mass is in particles that are larger than 1 μm.

For comparison, Fig. 12 contains similar data for a Teflon run where little difference is evident between the two DustTrak instruments, indicating that the bulk of the aerosol mass is in particles below 1 μm. Figure 13 shows that the effect of is a substantial decrease in the number concentration (Ptrak) while the mass concentrations (DustTraks) remain relatively steady. This behavior is consistent with particle coagulation with limited wall loss. It is also interesting to compare the response of the ISS and STS detectors for silicone rubber and Teflon. In figure 11, both the ISS detector and the STS detectors show strong signals. (Note that the presence of smoke yields a positive deflection of the ISS signal and a negative deflection on the STS signal). This is reasonable because the silicone rubber smoke has a larger diameter of average mass in Table 1. In figure 12, the STS detector provided a strong signal, whereas the signal from the ISS detector was more erratic, because the Teflon smoke is comprised primarily of particles having diameters less than 1 μm, as mentioned above. This shows that the smoke detectors used in STS and ISS have different response to particle size range.

Results from the moment instruments for baseline runs (no aging) and 720 second aging runs are summarized in Table 1.

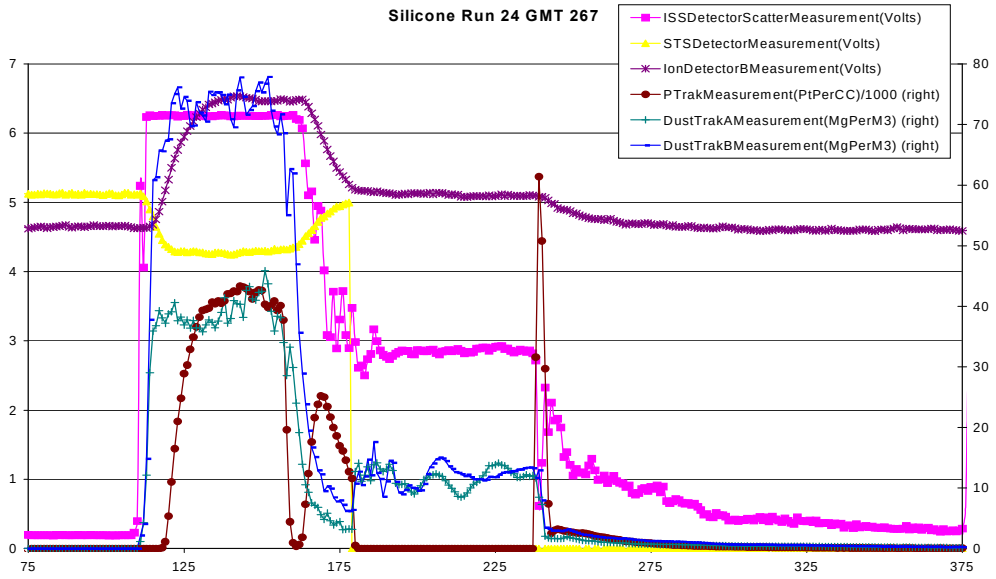


Figure 11. Instrument traces for a typical silicone rubber run. Base levels are at the beginning and end of the trace. The actual sampling interval is between 125 and 160 seconds. The differences in the rise times of each instrument are due to the smoke transit time to the instrument. The large difference between DustTrak A and DustTrak B is indicative of the substantial portion of the particle distribution that is larger than 1µm (the cutoff for the impactor on DustTrak A).

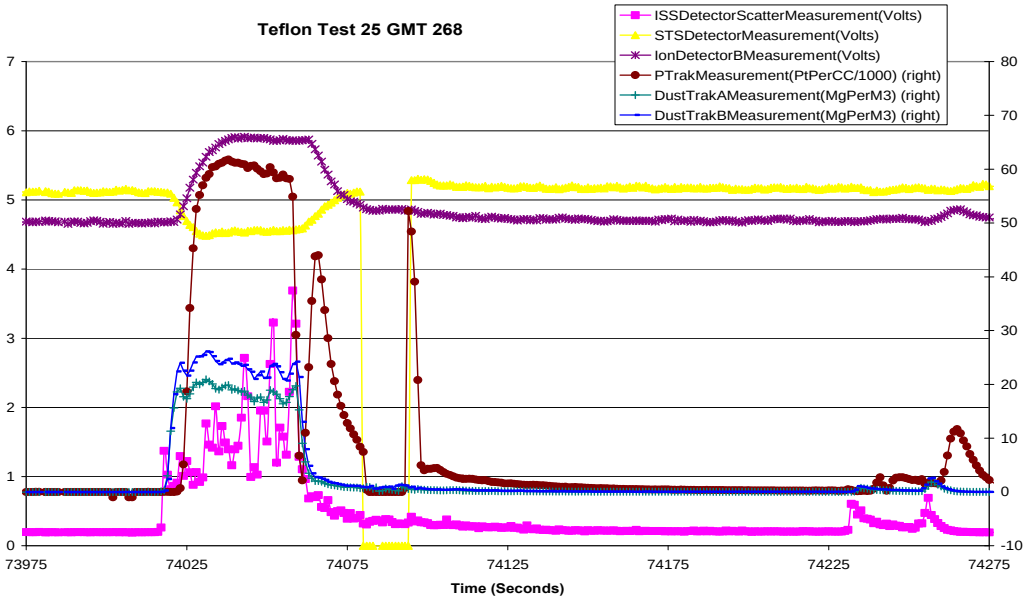


Figure 12. Instrument traces for a typical Teflon run. Base levels are at the beginning and end of the trace. The actual sampling interval is between 74025 and 74060 seconds. The small difference between DustTrak A and DustTrak B is indicative that the most of the particle distribution is smaller than 1µm (the cutoff for the impactor on DustTrak A).

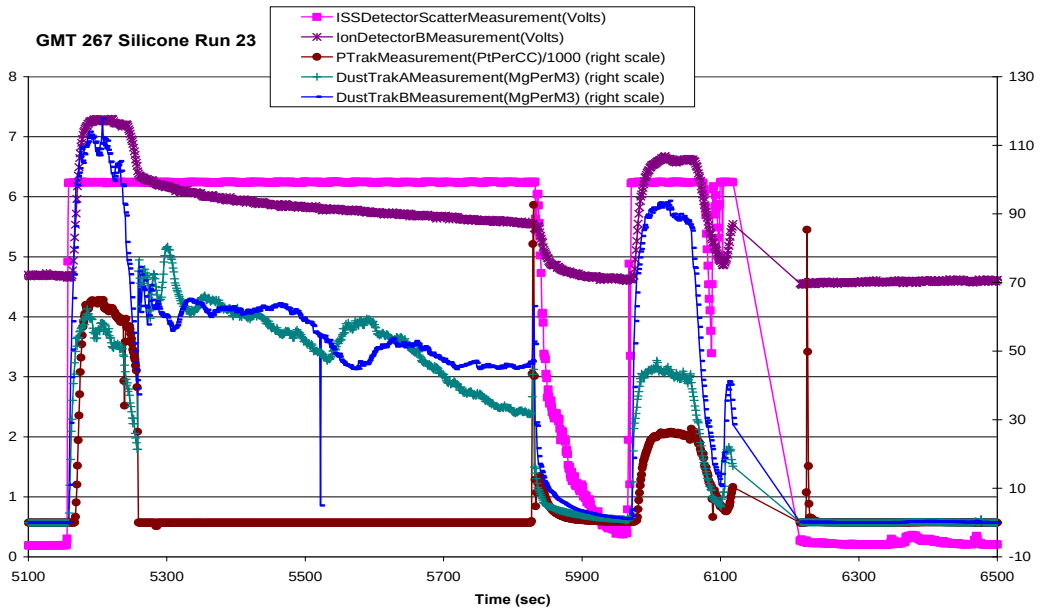


Figure 13. Instrument traces for a typical smoke aging run for silicone rubber. Base levels are at the beginning and end of the trace. The first sampling interval is between 5200 and 5250 seconds and the post aging sampling interval is between 6000 and 6050 seconds. Consistent with particle coagulation with limited wall loss, the overall number count (Ptrak) decreases substantially while the mass concentrations (DustTraks) remain relatively steady.

Table 1. Results of SAME flight tests

	Geom.Mean Diam. (nm)	Arithmetic Mean Diam. (nm)	Diam. of Average Mass (nm)	σ_g
Kapton	80	105	179	2.08
Kapton aged	155	186	267	1.82
Lamp wick	186	227	341	1.89
Lamp wick aged	258	308	438	1.81
Silicone	189	257	472	2.18
Silicone aged	352	422	606	1.83
Teflon	170	205	297	1.84
Teflon aged	228	273	392	1.82

The Teflon and Kapton particles are very small with Teflon showing limited increase in the size from the diameter of average mass compared to the arithmetic mean diameter. The lamp wick results exhibit substantially larger diameter of average mass. As reported previously,[12] the DustTrak results for lamp wick and silicone demonstrated substantial portions of the particle distribution possessed particle diameters larger than 1 μm. The small diameters for the Kapton and Teflon will make detection of this smoke challenging for light scattering devices, on the other hand the large sizes seen with the lamp wick and silicone would generate very large signals on a light scattering system, suggesting that detection of these particles against the background environment will require a detection system capable of measuring more than one moment of the particle size distribution. The aging results show substantial increase in the average particle sizes with them rising to over 1 μm for the aged silicone diameter of average mass. Since the aging can be considered to be representative of smoke emerging from a closed volume (e.g. an

avionics enclosure), fire sensors must consider large particles in addition to the smaller particles seen with Kapton.

In general, since the mass increases with the cube of the diameter, the larger particles do not affect the count mean diameter (CMD) as much as they affect the light scattering signal which scales with the particle mass. Consequently, although the CMD for silicone rubber is 189 nm, almost half the particle mass is larger than 1 μm, as evidenced by the ratio of the DustTrak instruments. The lamp wick shows similar behavior.

On the other hand, the Kapton and Teflon show substantially narrower particle size distributions with little mass in particles larger than 1μm. These low populations of larger particles are consistent with the less effective detection by light scattering devices which was seen for some of the Teflon tests and with the signals obtained from the STS and ISS detectors. All of the tests conducted in this series of experiments show strong signals on the Space Shuttle detector (ionization) despite the larger particle size distributions seen for the lamp wick (cellulose) and silicone rubber. The CSD experiment[5] reported poor detection by the Space Shuttle detector for non-flaming overheat of cellulose and silicone rubber (indicating larger particle sizes). The reduced signal on the Space Shuttle detector for CSD suggests that under some reasonable pre-fire conditions the smoke size distribution can extend to diameters beyond those seen in the SAME experiment.

The TEM grids recovered from the thermal precipitators were unfortunately contaminated with extraneous particles whose origin was believed to be somewhere in the assembly of the thermal precipitators. Despite the contamination, the lamp wick and Teflon samples produced distinct particles consistent with ground based experience. Typical particles are shown in Fig. 14.

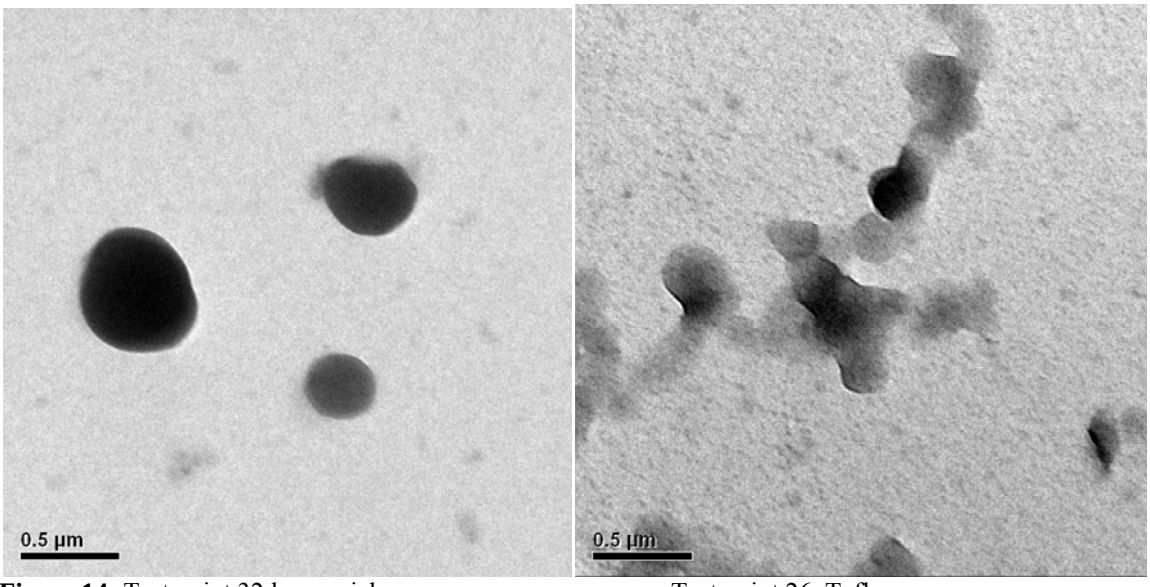


Figure 14: Test point 32 lamp wick Test point 26, Teflon

Several conference papers were prepared during the analysis of the data obtained from SAME and are included in the appendices. These provide additional details on several intermediate steps in the data analysis process that led to the final results described in this report.

6. Closing Remarks

The SAME project successfully implemented the moment method described in this report, under the volume constraints of the ISS MSG, to determine the size distribution of the smokes generated from 5 tested materials in well controlled conditions of heating temperature, heating time and air flow rate. In combination with the knowledge gained from previous CSD experiment, it becomes obvious that the detection of smoke across the size range seen in SAME and CSD, while still discriminating against dust (which tends to be large), will probably require a detection system that can discriminate the particle size distribution to at least a limited degree. Two possible effective means of achieving this size discrimination are to employ a detector that measures more than one moment of the size distribution or to use a detector that can classify the particles into size ranges.

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7. Moment Method for Estimating Micro-gravity Smoke Particle Size Parameters, Thomas Cleary, Claudia Rexfort, George Mulholland, David Urban, Jiann Yang, and Zeng-Guang Yuan, AAAR meeting Sept 10, 2006 St Paul Minn.
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Appendix A Log of SAME flight tests

Test Number	GMT	Sample Type	Reheat/Initial	Carousel	Number (Position)	Record Time	Comments
1	252	Silicone	Initial	1	1	11:50 - 13:20	
2	254	DBP	Initial	1	5	5:50 - 7:01	
3	254	DBP	Reheat	1	5	4:48 - 6:10	
4	255	DBP	Reheat	1	5	37:42 - 38:51	
5	255	Silicone	Initial	1	6	39:99 - 41:15	
6	256	Teflon	Initial	1	2		
7	256	Lampwick	Initial	1	3	18:54 - 20.25	
8	257	Kapton	Initial	1	4	1:13 - 1:46	run abort because of high Rt/Ro, finished manually
9	257	DBP	Reheat	1	5	1:13 - 2:50	
	257					3:40 -3:50	engage carousel
10	257	Teflon	Reheat	1	2	6:35 - 8:17	
	257					8:30 - 8:39	disengage carousel
11	262	DBP	Initial	2	4	1:39 - 3:07	
12	262	DBP	Reheat	2	4	5:37 - 7:05	
13	262	Teflon	Initial	2	1	9:36 - 10:59	
14	263	Kapton	Initial	2	3	33:05 - 34:18	
15	263	DBP	Initial	2	5	38:29 - 40:03	
16	263	DBP	Reheat	2	4	35:42 - 36:48	
17	264	Lampwick	Initial	2	2	14:05 - 15: 50	
	264					16:31 - 17:30	rotate carousel
18	264	DBP	Initial	2	6	<6.:39	
19	264	DBP	Reheat	2	6	6:39 - 8:00	
20	264	DBP	Reheat	2	6	8:00 - 11:00	
21	264	DBP	Reheat	2	6	19:32 - 21:10	
22	264	DBP	Reheat	2	6	22:30 - 24:16	
23	267	Silicone	Initial	3	1	12:28 - 13:48	
24	267	Silicone	Initial	3	2	17:40 - 19:10	
25	268	Teflon	Initial	3	4	17:58 - 19:12	
26	268	Teflon	Initial	3	5	20:33 - 21:58	
27	269	Silicone	Initial	3	3	1:48 - 3:09	
28	269	Teflon	Initial	3	6	4:35 - 5:36	
29	271	Kapton	Initial	4	4	1:22 - 2:23	
30	272	Kapton	Initial	4	5	1:00 - 2.30	
31	272	Lampwick	Initial	4	1	2:50 - 4:20	
32	272	Lampwick	Initial	4	2	8:35 - 9:57	
33	273	Kapton	Initial	4	6	9:58 - 10:59	missed first part of heating, using 2 cm/s
34	273	Lampwick	Initial	4	3	14:38 - 15:48	using 2 cm/s
						16: 18 - 16:27	disengage carousel
35	276	Silicone	Initial	5	1	10:10 - 11:05	
36	276	DBP	Initial	5	6	12:08 - 13:32	
37	276	DBP	Reheat	5	6	14:46 - 15:45	8 cm/s
38	276	DBP	Reheat	5	6	18:46 - 20:18	
39	276	DBP	Reheat	5	6	20:25 -21:55	
40	276	DBP	Reheat	5	6	22:58 - 23:58	
41	277	Teflon	Initial	5	2	3:57 - 5:15	
42	277	DBP	Initial	5	5	00:53 - 2.21	
43	277	DBP	Reheat	5	5	6:19 - 7:55	
44	277	DBP	Reheat	5	5	8:05 - 9:42	30 minute aging, 75 second heating
45	278	DBP	Reheat	5	5	10:58 - 12:17	
46	278	Lampwick	Initial	5	3	13:34 - 14:29	
47	278	Kapton	Intiial	5	4	15: 59 - 17:06	

2009-01-2468

Smoke Particle Sizes in Low-Gravity and Implications for Spacecraft Smoke Detector Design

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ABSTRACT

This paper presents results from a smoke detection experiment entitled Smoke Aerosol Measurement Experiment (SAME) which was conducted in the Microgravity Science Glovebox on the International Space Station (ISS) during Expedition 15. Five different materials representative of those found in spacecraft were pyrolyzed at temperatures below the ignition point with conditions controlled to provide repeatable sample surface temperatures and air flow conditions. The sample materials were Teflon®, Kapton®, cellulose, silicone rubber and dibutylphthalate. The transport time from the smoke source to the detector was simulated by holding the smoke in an aging chamber for times ranging from 10 to1800 seconds. Smoke particle samples were collected on Transmission Electron Microscope (TEM) grids for post-flight analysis. The smoke particle size distributions were measured using three particulate aerosol diagnostics that measure different moments of the size distribution and smoke detectors from the ISS and the space shuttle. These diagnostics were used to estimate the particle number concentration (zeroth moment), the diameter concentration (first moment) and the mass concentration (third moment). These values were combined to estimate the diameter of average mass, the count mean diameter, the geometric mean diameter and the geometric standard deviation. Even though the analysis of these data is ongoing, information from the various instruments and the calculated mean particle sizes has important implications for the design of spacecraft smoke detectors. These implications are reviewed in this paper relative to the operating principles of existing smoke detection technology. These results can be used to establish design requirements for smoke detectors for future spacecraft.

INTRODUCTION

Design of fire detector systems requires understanding of the background conditions and the expected range of events to be detected. Detectors for terrestrial fires were developed using the results of extensive study of terrestrial fires.[1,2] Unfortunately, there is no corresponding data set for spacecraft fires and consequently current spacecraft fire detectors were developed based upon experience from terrestrial systems. Given the large changes in fire behavior seen in reduced gravity, it is reasonable to expect that the smoke characteristics and detection requirements of these fires will change. The absence of buoyant flow in

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low-gravity increases the residence time in microgravity fires and increases the transit time from the reaction zone to the detector.[3] The materials used in spacecraft are different from typical terrestrial applications where smoke properties were previously evaluated. Very significantly, the limited options available to respond to a spacecraft fire increase the importance of detecting the fire in an earlier stage of its development than is typical for terrestrial applications. All of these effects can be expected to change the detection requirements: both the smoke concentration sensitivity, and the expected smoke particle size distribution. The objective of this work was to make sufficient measurements of smoke from spacecraft fires to enable improved design of future detectors.

SMOKE BACKGROUND

Smoke is a general term that encompasses aerosol materials produced by a number of processes. In particular it can include unburned, recondensed, original polymer or pyrolysis products that can be liquid, solid, carbonaceous soot, condensed water vapor, or ash particles. Soot particles dominate the smoke particulate in established flaming fires while unburned pyrolysis products and recondensed polymer fragments are produced by smoldering and pyrolysis in the early stage of fire growth. Given the constrained space on any spacecraft, the target for the fire detection system is necessarily the early phase and not established flaming fires; consequently, the primary target for detection is the pyrolysis products and not the soot.

Prior spacecraft systems are summarized in more detail in papers by Friedman and Urban.[4,5] In the Mercury, Gemini and Apollo missions, the crew quarters were limited and mission durations were short. Consequently it was considered reasonable that the astronauts would rapidly detect any fire. The Skylab module, however, included approximately 30 UV-sensing fire detectors.[4] These devices were limited to line-of-sight and were reported to have difficulties with false alarms. The Space Shuttle (Space Transportation System (STS)) detectors were based upon ionization fire detector technology, the most advanced technology available at the time and used an inertial separator designed to eliminate particles larger than 1-2 micrometers. The International Space Station (ISS) smoke detectors use near-IR forward scattering, rendering them most sensitive to particles larger than a micrometer, outside of the range of sensitivity of the shuttle detector.

More details of the ISS and STS detector requirements are presented by Steisslinger *et al*. [6] however the basic details are summarized below. The STS detector, as built, was designed to alarm at 2 mg/m^3 (based on 1 micrometer particles) or 0.022 $mg/m^3/s$ rise in concentration for 20 seconds. The ISS detector was designed to alarm at obscuration of 1% per foot using an Underwriters Laboratory (UL) smoke box and a white light extinction meter. This was implemented using a transfer standard detector and a 0.5 micrometer polystyrene latex-bead aerosol system that was used to set the amplifiers on each unit. The transfer standard was calibrated in the smoke box and then used to set the levels with the aerosol system.

As described by Friedman[4] there have been six overheat and failed component failures in the NASA Orbiter fleet in addition to several similar incidents that have occurred on the ISS. None of these events spread into a real fire but as mission durations increase, the likelihood of failures increases. The experience on Mir in 1997 has shown that failure of oxygen generation systems can have significant consequences. As a result, improved understanding of spacecraft fire detection is critically needed.[7]

Previous work on smoke particles from low-gravity sources by Urban *et al*.[5] found that the particulate produced by low-gravity flames (soot or unburned fuel particles) tends to have larger size particles than in normal gravity. Results from the CSD (Comparative Soot Diagnostics) Experiment[5] which studied smoke properties in low-gravity from several spacecraft materials suggested that liquid smoke particles could achieve sizes larger than 1 μm while solid particulate remained in the sub-micrometer range. However, the CSD experiment did not produce sufficient data concerning the size of the liquid smoke particles to guide detector design. The combined impact of these limited results and theoretical predictions is that, as opposed to extrapolation from 1-g data, direct knowledge of low-g combustion particulate is needed for more confident design of smoke detectors for spacecraft.

SAME EXPERIMENT CONCEPT

To address the limited data from the CSD experiment concerning the likely size of spacecraft smoke particulate, another experiment, the Smoke Aerosol Measurement Experiment (SAME), was developed. The SAME experiment sought to avoid the problems experienced by the CSD experiment by obtaining the particulate size statistics on-orbit with a reduced dependence upon sample return to Earth. This is a challenging endeavor because existing aerosol instrumentation is typically large, incompatible with spacecraft experiment constraints, and may require substantial sample return to Earth. As will be described below, an alternative approach was employed that used three discrete instruments to estimate separate moments of the size distribution. When combined, these moments provide useful aggregate statistics of the size distribution. The measurements were made using smoke generated by overheated spacecraft materials in much the same manner as the CSD experiment, however, the sample temperature, flow field, and particle aging time were more rigorously controlled.

MOMENT METHOD

The approach used by the SAME experiment is termed the 'moment method' for convenience.[8] As will be described below, the approach consists of measuring three moments of the size distribution (zeroth, first and third) and using the properties of the log-normal distribution to estimate the geometric mean diameter (GMD) and the geometric standard deviation (GSD).

The average particle size and the width of the size distribution will be estimated from various moments of the size distribution. The number distribution, $f_N(D)$, is defined as

$$f_N(D) = \frac{dN}{dD} \quad (1)$$

where dN is the number of particles per cm^3 with diameter between D and D + dD. The moments of interest consist of the number concentration, M_0, the first moment M_1, and the volume or mass concentration moment, M_3 and are defined as

$$M_i = \int D^i f_N(D) dD \quad i = 0,1,3 \qquad (2)$$

When i=0, the zeroth moment of the distribution, M_0, equation (2) is simply the number of particles per unit volume. In the SAME experiment, this was measured using a condensation nuclei counter. The first moment, i=1, can also be thought of as the "diameter concentration" or integrated diameter per unit volume and is approximately proportional to the ionization detector moment (signal). For particles in the Mie scattering regime, particles sizes from 0.3λ to about 3λ (~0.2 μm to 2.0 μm for a red laser), the light scattering signal is approximately proportional to the third moment, i=3. From these moments, and a measurement of M_0 using a condensation particle counter, two mean diameters can be computed: the count (arithmetic) mean diameter $D_{0.5}$ or $\overline{d}$, which is equal to M_1/M_0 and the diameter of average mass $D_{1.5}$ or $d_{\overline{m}}$, which is equal to $(3/(4\pi\rho)M_3/M_0)^{1/3}$. The basis for the subscript naming convention for $D_{0.5}$ and $D_{1.5}$ will be discussed later). The log-normal size distribution is widely used for describing the size distribution of aerosols including non-flaming smoke because, for most aerosols, the bulk of the number concentration is associated with smaller particles.[9, 10] The number distribution $f_N(D)$ for the lognormal distribution is expressed as follows:

$$f_N(D) = \frac{N_t}{(2\pi)^{1/2} D \ln\sigma_g} \exp\left(-\frac{(\ln D - \ln D_g)^2}{2\ln^2\sigma_g}\right) \quad (3)$$

where N_t is the total number concentration of the aerosol (=M_0), and D_g and σ_g are the geometric mean diameter and geometric standard deviation defined by

$$\ln D_g = \int_0^\infty \ln D f_N(D) dD \Big/ \int_0^\infty f_N dD \qquad (4)$$

$$\ln\sigma_g = \left[\int_0^\infty (\ln D - \ln D_g)^2 f_N(D) dD \Big/ \int_0^\infty f_N(D) dD\right]^{1/2} \quad (5)$$

For the log-normal distribution, one finds that the various diameter definitions given above are related to the GMD, D_g, via the equation:[9,10]

$$D_p = D_g \exp(p \ln^2 \sigma_g) \qquad (6)$$

For the count mean diameter, $D_{0.5}$, and the diameter of average mass, $D_{1.5}$, the corresponding values of p are 0.5 and 1.5. Fig. 1 shows a typical log-normal distribution for a D_g=1.0 μm and σ_g= 1.6. For this distribution, the corresponding values of $D_{0.5}$ and $D_{1.5}$ are1.17 μm and 1.39 μm, respectively. Using equation (6), one can relate σ_g to the ratio of $D_{1.5}$ and $D_{0.5}$ via the equation:

$$\sigma_g = \exp\left(\ln(D_{1.5}/D_{0.5})\right)^{1/2} \qquad (7)$$

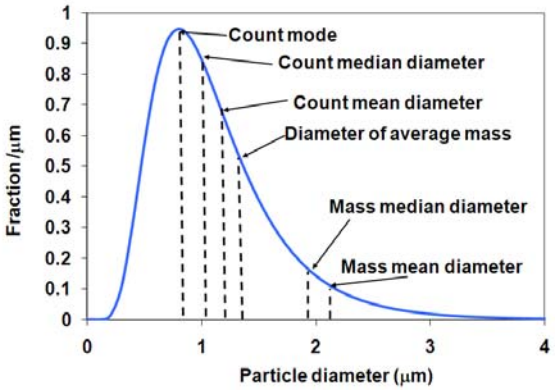


Figure 1: Diameters for a log-normal distribution with D_g=1.0 and σ_g= 1.6.

By combining these three moments it is possible to compute three mean diameters of the size distribution and the geometric standard deviation. Validation of this approach is discussed in Cleary, Weinert and Mulholland.[8] These statistics provide a strong basis for design of spacecraft smoke detectors.

INSTRUMENTS

More details of the experiment hardware and its calibration are presented elsewhere,[11,12] however a brief summary follows. The moment measurements were made using an assembly of three separate instruments. Two are industrial hygiene instruments manufactured by TSI and one is a modified residential smoke detector.

The zeroth-moment instrument is a condensation nuclei counter P-Trak® (TSI Inc.). This device operates by passing the aerosol-laden particle stream through a region saturated with isopropanol vapor and then into a cooler region where the vapor condenses onto the particles increasing their diameter such that they can be readily counted by a light scattering detector. This instrument is very robust and operates over a range of 0 to 10^5 particles/cm^3 and 20 nm to 4 to 5 μm diameter. Some dilution is required, since the allowable smoke concentration ranges from about 0.5×10^6 to 5×10^6 particles / cm^3.

The first-moment instrument is an ionization chamber from a residential smoke detector. This device uses an alpha-particle emitter (Am^{241}) to generate ions in a region within a DC electric field. The drift of the ions in the electric field results in a current. The presence of particulate reduces the current as a result of the attachment of the ions to the particulate. The mobility of the charged aerosol is too small for it to be collected on the ionization chamber electrode. The required particle concentrations are on the order of 10^5 particles/cm^3, and no sample dilution was required.

The third-moment instrument is a light scattering device DustTrak® (TSI Inc.). The device measures the intensity of light scattered by the smoke aerosol at a scattering angle of about 90 degrees light scattering signal to quantify the aerosol mass. This light scattering signal is approximately proportional to the mass concentration; however, to compensate for the non-linearity of the relationship between mass and the scattering signal density, additional analysis based on Mie Scattering theory or calibration with the specific aerosol material is required. The device's operating range is from 0.001 mg/m^3 to 100 mg/m^3. These devices are equipped with an aerodynamic impactor at the inlet which captures particles larger than the selected size. The SAME experiment included 2 DustTraks® one with a 1 μm impactor and one with a 10 μm impactor. The difference in the signal from these two devices provided a measure of the fraction of the particulate that was larger than 1 μm. In some cases dilution was required owing to the high smoke concentration levels.

A schematic of the assembled hardware appears in Fig. 2. The system was installed in the Microgravity Science Glovebox, an ISS facility that provides many resources including: containment, power, data, video and uplink commanding. Smoke was generated by overheating a small sample of material in the smoke generation duct for approximately 60 seconds. During this interval, controlled flow was induced by a moving piston in the aging chamber which drew the smoke into the chamber. The smoke was held in the aging chamber for a predetermined time, allowing the particles to coagulate. After a specified aging time, the smoke was then pushed by the piston into the diagnostics duct where the moment instruments made their measurements. Also installed in the diagnostics duct were space shuttle and ISS smoke detectors. As the smoke was monitored by the moment instruments, a sample of the smoke particles was deposited on a Transmission Electron Microscope (TEM) grid via a thermal precipitator which uses thermophoresis to deposit the particles on the grids.

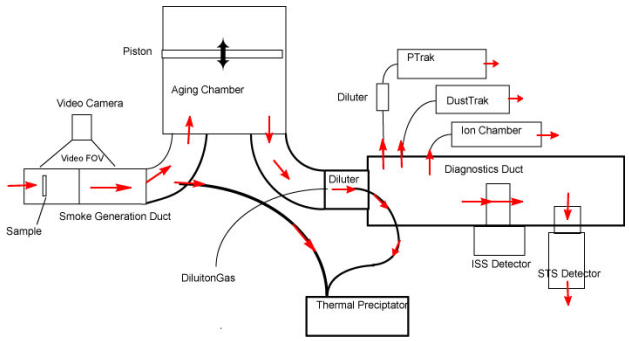


Figure 2. **Schematic of the SAME hardware**.

The zeroth and first moment devices showed good linearity over the conditions of interest however, the third moment device (TSI- DustTrak®) is theoretically predicted to show non-monotonic behavior as particle size is increased. While a satisfactory prediction of the DustTrak® response was obtained using liquid droplets of dioctyl phthalate, the uncertainty associated with the Mie scattering calculations due to the particle refractive index estimate, the potential of agglomerated particles, and deviation from assumed log-normal size distribution must be assessed further for the smoke samples here. In the results reported here, the DustTrak® response was directly calibrated, for each smoke source, against mass concentration measurements using a tapered element oscillating microbalance. This approach is valid as long as the size distribution of the sample does not change significantly. Using estimated material densities, the diameter of average mass was computed for each sample. Equation 7 was used to compute the geometric standard deviation from the arithmetic mean diameter and the diameter of average mass, and Equation 6 was used to compute the GMD.

EXPERIMENTAL RESULTS

Overall, 30 sample materials were tested. These were comprised of six samples each of 5 materials: Teflon®, Kapton®, silicone rubber, cellulose (lamp wick), and dibutyl-phthalate deposited on a porous wick. Geometric Mean (GMD), Arithmetic Mean, and Average Mass diameters and the Geometric Standard Deviation (GSD) results from the moment instruments for baseline runs (no aging) and 720 second aging are presented in Table 1.

The Teflon® and Kapton® particles were very small with Teflon® showing limited increase in the size from the diameter of average mass compared to the arithmetic mean diameter. The lamp wick results exhibited substantially larger diameter of average mass. As reported previously,[11] the DustTrak results for lamp wick and silicone rubber demonstrated that substantial portions of the particle distribution possessed aerodynamic diameters larger than 1 micrometer. The small diameters for the Kapton® and Teflon® will make detection of this smoke challenging for light scattering devices; on the other hand, the large sizes seen with the lamp wick and silicone rubber would generate very large signals on a light scattering system, suggesting that detection of these particles against the background environment will require a detection system capable of measuring more than one moment of the particle size distribution. The aging results showed substantial increase in the average particle sizes with the diameter of average mass rising to over 500 nm for the aged silicone rubber. Since the aging process can be considered to be representative of smoke emerging from a closed volume (e.g. an avionics enclosure), fire sensors must consider large particles in addition to the smaller particles seen with Kapton®.

	Geom. Mean Diam. (nm)	Arithmetic Mean Diam. (nm)	Diam. of Average Mass (nm)	σ_g
Kapton	80	105	179	2.08
Kapton aged	155	186	267	1.82
Lamp wick	186	227	341	1.89
Lamp wick aged	258	308	438	1.81
Silicone	189	257	472	2.18
Silicone aged	352	422	606	1.83
Teflon	170	205	297	1.84
Teflon aged	228	273	392	1.82

DETECTOR DESIGN REQUIREMENT IMPLICATIONS

To develop requirements from these results, it is instructive to consider some basic overarching criteria:

1. The requirement must not effectively define the technology
2. The requirement must be verifiable in a laboratory.
3. If the requirement describes or implies a test that cannot be done on every unit, it must be possible to define a subtest that can be conducted to qualify production units.
4. The requirement should represent the best understanding of spacecraft smoke conditions.

One possible method to meet these criteria is to develop a requirement that states an aerosol condition at alarm. To provide some perspective on the impact of the choice of such requirements, Figs. 3 and 4 present the outcome of a hypothetical single point requirement if it were implemented with different detector technologies. Each of these technologies respond to different particle characteristics such as the first or third moment of the distribution or the scattering signal. These figures were developed assuming the alarm level was set at 2 mg/m^3 for a log-normal particle size distribution with a 500 nm geometric mean diameter (GMD) and a geometric standard deviation (GSD) of 1.3. The values for the moment devices were calculated using Eq. 6 and the values for the scattering device were calculated using the Mie scattering model of Gockel.[13] In practice, a fire could produce a smoke particle size distribution with a GMD different than 500 nm yet each of these detectors would have an alarm point determined by their response at the alarm-requirement aerosol. For each type of detector, the alarm number concentration (Fig. 3) or alarm mass concentration (Fig. 4) were calculated as a function of the particle diameter. This is the concentration needed to achieve the same signal level as the alarm-requirement aerosol (2 mg/m^3 mass concentration, 500 nm GMD, and 1.3 GSD). As is evident in the graphs, depending upon the selected instrument, strong deviations are seen in number concentration and mass concentration at either the small or large ends of the particle distribution. For example, as particle diameter is increased, progressively smaller number concentrations are needed to reach alarm for

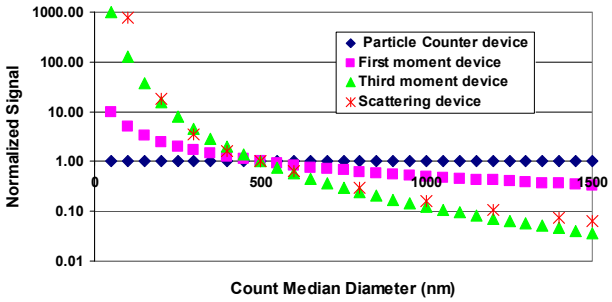


Figure 3: Normalized Number concentration at alarm for 4 smoke detector technologies assuming the alarm level was set at 2 mg/m^3 for 500 nm GMD and a geometric standard deviation of 1.3.

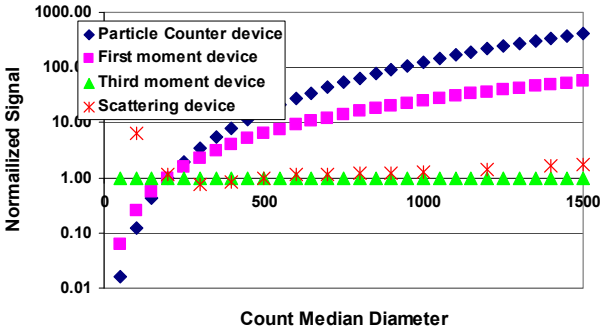


Figure 4: Normalized mass concentration at alarm for 4 smoke detector technologies assuming the alarm level was set at 2 mg/m^3 for 500 nm GMD and a geometric standard deviation of 1.3.

the scattering detector. This could indicate increased sensitively to nuisance sources such as dust. On the other hand, the number concentrations for alarm also rise quite high at smaller sizes for the same detector. Similar dependencies are seen with all detectors suggesting that a detector that measures a single moment of the size distribution may suffer from conditions where it is either too sensitive or not sensitive enough. Consequently a specification for a single particle size (or no size specification) would be risky. If the requirement is limited to a single particle size, the result can easily be a detector that is insensitive to the other sizes. This is essentially the case for the current ISS and STS detectors which are tuned to different size ranges and have very different sensitivities to particle size.[5] An alternative is to develop alarm requirements that include criteria at particle sizes spanning the expected range of fire particulate (e.g. 200 to 1200 nm). A detector that involves two moment measurements may be able to capitalize on the size information that is generated by the ratio between the two moment values. As can be seen in Figs. 3 and 4, the systematic variation between the alarm signals of different instruments could be used to discern size information This is similar to the approach suggested by Litton *et al.* who demonstrated that by combining two simple particle sensors that measure different moments, an inexpensive assessment of the particle size could be achieved.[14] With a detector that provides a combination of a concentration or moment measurement and an indication of the size, it would be possible to satisfy detection requirements that include concentrations at more than one particle size and the two signals might also help reduce false alarms

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ACRONYNMS

CSD: Comparative Soot Diagnostics

DAFT: Dust Aerosol measurement Feasibility Test

GMD: Geometric Mean Diameter

GSD: Geometric Standard Deviation

ISS: International Space Station

MSG: Microgravity Science Glovebox

SAME: Smoke Aerosol Measurement Experiment

STS: Space Transportation System

TEM: Transmission Electron Microscope

UL: Underwriters Laboratory

Properties of Smoke from Overheated Materials in Low-Gravity

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Smoke particle size measurements were obtained under low-gravity conditions by overheating several materials typical of those found in spacecraft. The measurements included integral measurements of the smoke particles and physical sample of the particles for Transmission Electron Microscope analysis. The integral moments were combined to obtain geometric mean particle sizes and geometric standard deviations. These results are presented with the details of the instrument calibrations. The experimental results show that, for the materials tested, a substantial portion of the smoke particles are below 500 nm in diameter.

I. Introduction

Appropriate design of fire detection systems requires knowledge of both the expected signature of the events to be detected and the background levels of the measured parameters. Terrestrial fire detection systems have been developed based on extensive study of terrestrial fires.[1, 2] Unfortunately there is no corresponding data set for spacecraft fires and consequently the fire detectors in current spacecraft were developed based upon terrestrial designs. There are a number of factors that can be expected to affect the particle size distribution of the smoke from spacecraft fires. The absence of buoyant flow in low-gravity increases the residence time in microgravity fires and increases the transit time from the reaction zone to the detector.[3] Microgravity fires have been found to have radically different structure from their 1-g counterparts. The limited options available to respond to a spacecraft fire increase the importance of early detection. Finally the materials used in spacecraft are different from typical terrestrial applications where smoke properties were previously evaluated. All of these effects can be expected to change the smoke particle size distribution. The objective of this work was to make sufficient measurements of smoke from spacecraft fires to enable improved design of future detectors.

Smoke Background

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Smoke is a general term that encompasses aerosol materials produced by a number of processes. In particular it can include unburned, recondensed, original polymer or pyrolysis products that can be either liquid or solid, hydrocarbon soot, condensed water vapor, and ash particles. Soot particles dominate the smoke particulate in established flaming fires while unburned pyrolysis products and recondensed polymer fragments are produced by smoldering and pyrolysis in the early stage of fire growth. Given the constrained space on any spacecraft, the target for the fire detection system is necessarily the early phase and not established flaming fires; consequently, the primary target for detection is the pyrolysis products and not the soot.

Prior spacecraft systems are summarized in more detail in papers by Friedman and Urban.[4, 5] In the Mercury, Gemini and Apollo missions, the crew quarters were limited and mission durations were short, consequently it was considered reasonable that the astronauts would rapidly detect any fire. The Skylab module, however, included approximately 30 UV-sensing fire detector.[4] These devices were limited to line-of-sight and were reported to have difficulties with false alarms. The Space Shuttle Detectors were based upon ionization fire detector technology, the most advanced technology available at the time and used an inertial separator designed to eliminate particles larger than 1-2 micrometers. The International Space Station (ISS) smoke detectors use near-IR forward scattering, rendering them most sensitive to particles larger than a micrometer, outside of the range of sensitivity of the shuttle detector. As described by Friedman[4] there have been six overheat and failed component failures in the NASA Orbiter fleet in addition to several similar incidents that have occurred on the ISS. None of these events spread into a real fire but as mission durations increase, the likelihood of failures increases. The experience on Mir in 1997 has shown that failure of oxygen generation systems can have significant consequences. As a result, improved understanding of spacecraft fire detection is critically needed.[6]

Previous work on smoke particles from low-gravity sources by Urban et al.[5] found that the particulate produced by low-gravity flames (soot or unburned fuel particles) tends to have larger size particles than in normal gravity. Results from the CSD (Comparative Soot Diagnostics) Experiment[5] which studied smoke properties in low-gravity from several spacecraft materials suggested that liquid smoke particles could achieve sizes larger than 1 μm while solid particulate remained in the sub-micrometer range. However, the CSD experiment did not produce sufficient data concerning the size of the liquid smoke particles to guide detector design. The combined impact of these limited results and theoretical predictions is that, as opposed to extrapolation from 1-g data, direct knowledge of low-g combustion particulate is needed for more confident design of smoke detectors for spacecraft.

II. SAME Experiment

To address the limited data from the CSD experiment concerning the likely size of spacecraft smoke particulate, another experiment the Smoke Aerosol Measurement Experiment (SAME) was developed. The SAME experiment sought to avoid the problems experienced by the CSD experiment by obtaining the particulate size statistics on-orbit with a reduced dependence upon sample return to Earth. This is a challenging endeavor because existing aerosol instrumentation is typically large, incompatible with spacecraft experiment constraints, and may require substantial sample return to Earth. As will be described below, an alternative approach was employed that used three discrete instruments to measure separate moments of the size distribution. When combined, these moments provide useful aggregate statistics of the size distribution. The measurements were made using smoke generated by overheated spacecraft materials in much the same manner as the CSD experiment however the sample temperature, flow field, and particle aging time were more rigorously controlled.

A. Moment Method

The approach used by the SAME experiment is termed the 'moment method' for convenience.[7] As will be described below, the approach consists of measuring three moments of the size distribution (zeroth, first and third) and using the properties of the log-normal distribution to estimate the geometric mean diameter and the standard deviation.

The average particle size and an estimate of the width of the size distribution will be estimated from various moments of the size distribution. The number distribution, $f_N(D)$, is defined as

$$f_N(D) = \frac{dN}{dD} \tag{1}$$

where dN is the number of particles per cm^3 with diameter between D and D + dD. The moments of interest consist of the number concentration, M_0, the first moment M_1, and the volume or mass concentration moment, M_3 and are defined as

$$M_i = \int D^i f_N(D) dD \quad i = 0,1,3 \tag{2}$$

When i=0, the zeroth moment of the distribution, M_0, equation (2) is simply the number of particles per unit volume. In the SAME experiment, this was measured using a condensation nuclei counter. The first moment, i=1, can also be thought of as the "diameter concentration" or integrated diameter per unit volume and is approximately proportional to the ionization detector moment (signal). For particles in the Mie scattering regime, particles sizes from 0.3λ to about 3λ (~0.2 μm to 2.0 μm for a red laser), the light scattering signal is approximately proportional to the third moment, i=3. From these moments, and a measurement of M_0 using a condensation particle counter, two mean diameters can be computed: the count (arithmetic) mean diameter $D_{0.5}$ or $\bar{d}$, which is equal to M_1/M_0 and the diameter of average mass $D_{1.5}$ or $\bar{d}_m$, which is equal to $(M_3/M_0)^{1/3}$. The basis for the subscript naming convention for $D_{0.5}$ and $D_{1.5}$ will be discussed later). The log-normal size distribution is widely used for describing the size distribution of aerosols including non-flaming smoke because for most aerosols; the bulk of the number concentration is associated with smaller particles.[8, 9] The number distribution fN(D) for the lognormal distribution is expressed as follows:

$$f_N(D) = \frac{N_t}{(2\pi)^{1/2} D \ln \sigma_g} \exp\left(- \frac{\left(\ln D - \ln D_g\right)^2}{2 \ln^2 \sigma_g} \right) \tag{3}$$

where N_t is the total number concentration of the aerosol (=M_0), and D_g and σ_g are the geometric mean diameter and geometric standard deviation defined by

$$\ln D_g = \int_0^\infty \ln D f_N(D) dD \Big/ \int_0^\infty f_N dD \tag{4}$$

$$\ln \sigma_g = \left[\int_0^\infty (\ln D - \ln D_g)^2 f_N(D) dD \Big/ \int_0^\infty f_N(D) dD \right]^{1/2} \tag{5}$$

For the log-normal distribution, one finds that the various diameter definitions given above are related to the geometric mean number diameter, D_g, via the equation:[8 9]

$$D_p = D_g \exp(p \ln^2 \sigma_g) \tag{6}$$

For the count mean diameter, $D_{0.5}$, and the diameter of average mass, $D_{1.5}$, the corresponding values of p are 0.5 and 1.5. Fig. 1 shows a typical log-normal distribution for a D_g=1.0 and σ_g= 1.6. For this distribution, the corresponding values of $D_{0.5}$ and $D_{1.5}$ are1.17 μm and 1.39 μm, respectively. Using equation (6), one can relate σ_g to the ratio of $D_{1.5}$ and $D_{0.5}$ via the equation:

$$\sigma_g = \exp\left(\ln(D_{1.5} / D_{0.5})\right)^{1/2} \tag{7}$$

By combining these three moments it is possible to compute three mean diameters of the size distribution and the geometric standard deviation. Validation of this approach is discussed in Cleary, Weinert and Mulholland.[7] These statistics provide a strong basis for design of spacecraft smoke detectors.

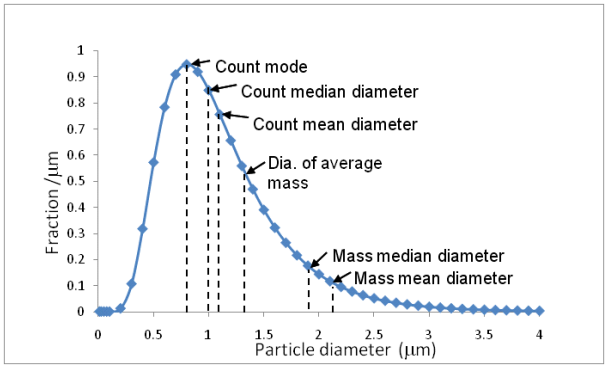


Figure 1: Diameters for a log-normal distribution with D_g=1.0 and σ_g= 1.6.

B. Instruments

These measurements were made using an assembly of three separate instruments. Two are industrial hygiene instruments manufactured by TSI and one is a modified residential smoke detector.

The zeroth-moment instrument is a condensation nuclei counter P-Trak™ (TSI Inc.). This device operates by passing the aerosol-laden particle stream through a region saturated with isopropanol vapor and then into a cooler region where the vapor condenses onto the particles increasing their diameter such that they can be readily counted by a light scattering device. This instrument is very robust and operates over a range of 0 to 10^5 particles/cm^3 and 20 nm to 1 µm diameter. Some dilution is required, since the smoke concentration ranges from about 0.5 × 10^6 to 5 × 10^6 particles / cm^3 There was also a concern that the isopropanol condensate would not return to the wick in low-gravity.[5] To mitigate this issue, the condensing section of the device was modified with very small grooves to improve conductance of the condensate back to the wick. These changes were tested in a separate space experiment with good results indicating the modified device could be used successfully in low gravity.[5]

The first-moment instrument is the ionization chamber from a residential smoke detector. This device uses an alpha-particle emitter to generate ions in a region within a DC electric field. The drift of the ions in the electric field results in a current. The presence of particulate reduces the current as a result of the attachment of the ions to the particulate. The mobility of the charged aerosol is too small for it to be collected on the ionization chamber electrode. The required particle concentrations are on the order of 10^5 particles/cm^3 and no sample dilution was required.

The third-moment instrument is a light scattering device DustTrak™ (TSI Inc.). The device uses a 90 degree light scattering signal to quantify the aerosol mass density. For terrestrial dust particulate this signal correlates well with the mass concentration, however additional compensation will be needed to account for the range of particle sizes that will be seen in the SAME experiment. The device's operating range is from 0.001 mg/m^3 to 100 mg /m^3. These devices are equipped with an aerodynamic impactor at the inlet which captures particles larger than the selected size. The SAME experiment included 2 DustTraks™ one with a 1 µm impactor and one with a 10 µm impactor. The difference in the signal from these two devices provided a measure of the fraction of the particulate that was larger than 1 µm. In some cases dilution was required owing to the high smoke concentration levels.

A schematic of the assembled hardware appears in Fig. 2. The system was installed in the Microgravity Science Glovebox, an ISS facility that provides many resources including: containment, power, data, video and uplink commanding. Smoke was generated by overheating a small sample of material in the smoke generation duct for approximately 60 seconds. During this interval, controlled flow was induced by a moving piston in the aging chamber which drew the smoke into the chamber. The smoke was held in the chamber for a predetermined time, allowing the particles to coagulate. After a specified aging time, the smoke was then pushed by the piston into the diagnostics duct where the moment instruments made their measurements. Also installed in the diagnostics

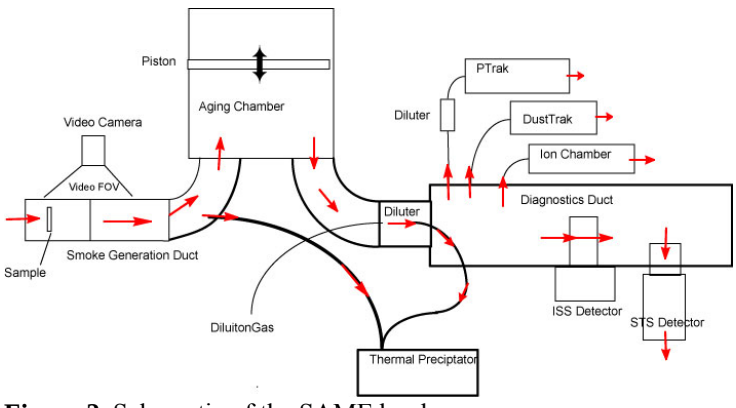


Figure 2. Schematic of the SAME hardware.

duct were space shuttle and ISS smoke detectors. As the smoke was monitored by the moment instruments, a sample of the smoke particles was deposited on Transmission Electron Microscope (TEM) grid via a thermal precipitator which uses thermophoresis to deposit the particles on the grids.

Calibration of these instruments was essential to properly interpret the flight data and was performed on the ground before the flight. Calibration was accomplished using two different aerosol generators one using mono-disperse particle generation using dioctyl phthalate (DOP) according to the approach by Mulholland and Liu[10] and the other using polystyrene spheres. The monodisperse droplet generator functioned by producing a spray of DOP diluted with isopropanol which is then evaporated and recondensed producing monodisperse droplets. The droplet size is controlled by the DOP dilution level. The generator will operate stably for tens of minutes. The aerosol from the generator was sampled simultaneously by the SAME instrument under test and a reference standard. For the number count, the reference instrument was a TSI 3022 particle counter, for the Mass Concentration, a Tapered Element Oscillating Microbalance and for the first moment a TSI Electrical Aerosol Detector was used. The results for the P-Trak are shown in Fig. 3. As the number concentration increased, the effect of the particle diameter became more evident. Separate correlations were developed for each particle size and the closest correlation was used to analyze the flight data based on the initial estimates of the average particle size.

The first moment device, the ion chamber showed little effect of particle size as seen in Fig 4. Consequently a single correlation was used for all particle sizes.

The third moment device (TSI-Dust Trak) is theoretically predicted to show non-monotonic behavior as particle size is increased. The calibration results for the DustTrak shown in Fig. 5 was obtained using monodisperse DOP particles and indicates a strong dependence on particle size. This dependence becomes even more complicated when the particle size distribution is polydisperse.

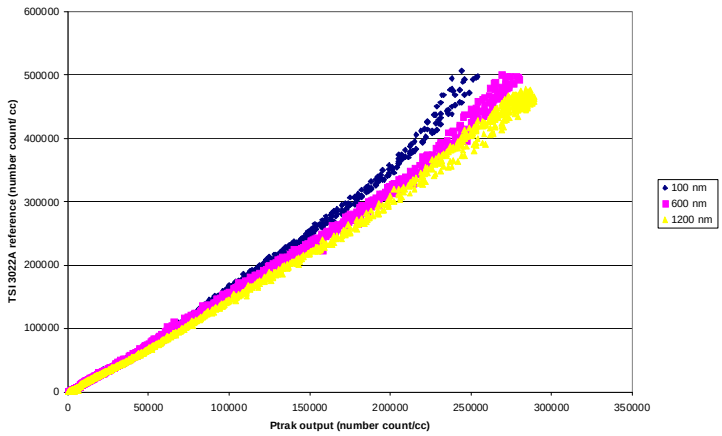


Figure 3: Ptrak Calibration results with Mono disperse DOP droplets.

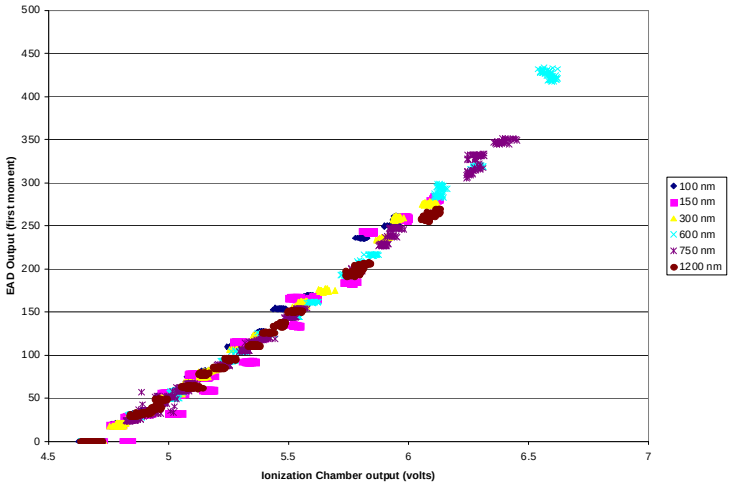


Figure 4: Ion Chamber Calibration results with Mono disperse DOP droplets.

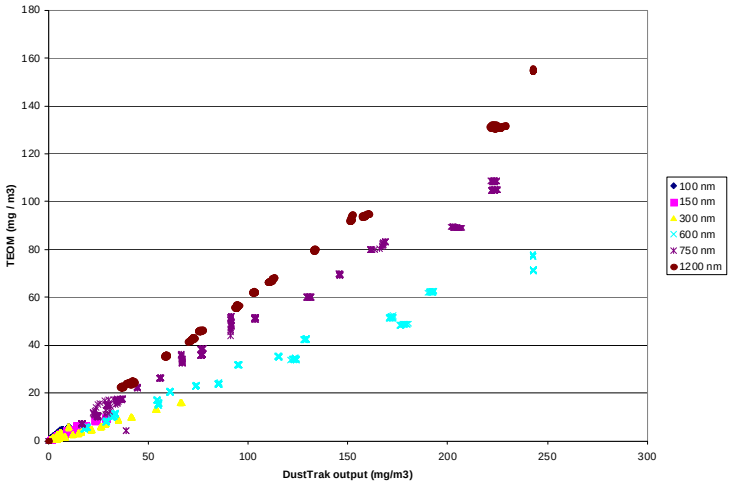


Figure 5: Dust Trak results with Mono disperse DOP droplets.

To address this calibration while minimizing the assumptions required about the particle size distribution, a model of the operation of the DustTrak instrument was developed using Mie scattering theory. Using the zeroth and first moment data along with the refractive index of the particulate, the third moment was predicted by assuming a log-normal distribution and making initial guesses for the geometric mean diameter and the geometric standard deviation. The output of the DustTrak was then predicted by the scattering model and the model proceeded in an iterative manner until the predicted DustTrak output matched the actual output. The geometric mean diameter and geometric standard deviation were those that produced convergence of the predicted and measured DustTrak output. In some cases, convergence was not obtained and for these cases, the geometric mean diameter can not be determined. This non-convergence is currently under investigation but may be due to the presence of a bimodal particle size distribution.

C. Experimental Results

Overall 30 sample materials were tested. These were comprised of six samples each of 5 materials: Teflon™, Kapton™, silicone rubber, cellulose (lamp wick), and dibutyl-phthalate deposited on a porous wick. The TEM girds recovered from the thermal precipitators were unfortunately contaminated with extraneous particles whose origin is currently under investigation. Despite the contamination, the lamp wick and Teflon samples produced distinct particles consistent with ground based experience. Typical particles are shown in Fig. 6 Geometric Mean, Arithmetic Mean, and Average Mass diameter results from the moment instruments for baseline runs are presented in Table 1.

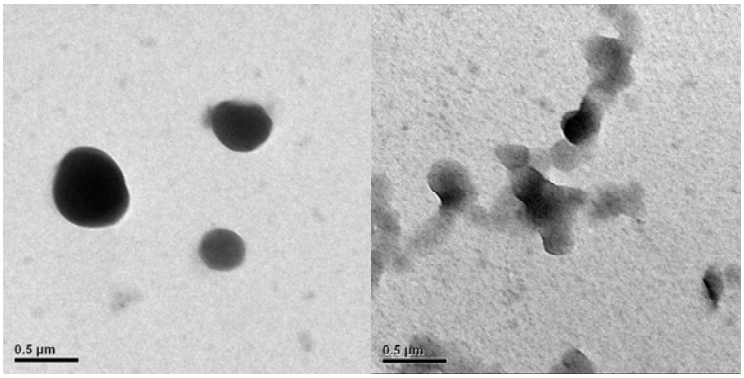


Figure 6: Test point 32 Lamp wick Test point 26, Teflon

	Geometric Mean Diameter (micro meters)	Arithmetic Mean Diameter (micro meters)	Diameter of Average Mass (micro meters)
Kapton	0.080	0.105	0.172
Lamp wick	0.186	0.227	0.420
Silicone	0.189	N/A	N/A
Teflon	0.159	0.205	0.219

Table 1:Diameter results from baseline runs. The Silicone results did not converge for these conditions

Overall, the Teflon and Kapton particles were very small with limited increase in the size from the diameter of average mass compared to the arithmetic mean diameter. The Lamp wick results exhibited substantially larger diameter of average mass. As reported previously,[11] the Dust Trak results for Lamp wick and Silicone demonstrated substantial portions of the particle distribution possessed aerodynamic diameters larger than 1 micrometer. The small diameters for the Kapton and Teflon will make detection of this smoke challenging for light scattering devices, on the other hand the large sizes seen with the Lamp wick and Silicone would generate very large signals on a light scattering system, suggesting that detection of these particles against the background environment will require a detection system capable of measuring more than one moment of the particle size distribution.

III. Acknowledgments

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Measurement of Smoke Particle Size under Low-Gravity Conditions

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ABSTRACT

Smoke detection experiments were conducted in the Microgravity Science Glovebox (MSG) on the International Space Station (ISS) during Expedition 15 in an experiment entitled Smoke Aerosol Measurement Experiment (SAME). The preliminary results from these experiments are presented. In order to simulate detection of a prefire overheated-material event, samples of five different materials were heated to temperatures below the ignition point. The smoke generation conditions were controlled to provide repeatable sample surface temperatures and air flow conditions. The smoke properties were measured using particulate aerosol diagnostics that measure different moments of the size distribution. These statistics were combined to determine the count mean diameter which can be used to describe the overall smoke distribution.

INTRODUCTION

Current spacecraft fire detection systems have all been designed based on normal gravity fire behavior and properties. This approach was necessary given the absence of data concerning microgravity smoke particulate. However planned spacecraft and associated missions outside of low earth-orbit will require increased reliability of fire detection systems in addition to robust false alarm resistance. Extended lunar-surface missions will require systems that are not influenced by lunar dust and other nuisance dust sources. The absence of buoyant flow increases the residence time in microgravity fires causing substantial changes in flame structure and consequently can change the properties of the produced smoke. In particular, the increased residence time in the high temperature zone where the smoke is formed, can increase the opportunity for growth in the size of the smoke particulate. Furthermore, the increased transit time to the detector seen in low-gravity smoke detection scenarios [1] provides additional time for growth in particle size through aging or low temperature agglomeration of smoke particles. The objective of this work is to make sufficient measurements of smoke from spacecraft fires to enable improved design of future detectors.

SMOKE BACKGROUND

The background in fire detection in US spacecraft is discussed by Urban et al. [2] however some details will be summarized here. Smoke is a general term that encompasses aerosol materials produced by a number of processes. In particular it can include unburned, recondensed, original polymer or pyrolysis products that can be either liquid or solid, hydrocarbon soot, condensed water vapor, and ash particles. Ash and soot particles dominate the smoke particulate in established 1-g flaming fires while unburned pyrolysis products and recondensed polymer fragments are produced by smoldering and pyrolysis in the early stage of fire growth. Given the constrained space on any spacecraft, the target for the fire detection system is necessarily the early phase and not established flaming fires; consequently, the primary target for detection is the pyrolysis products and not the soot and ash.

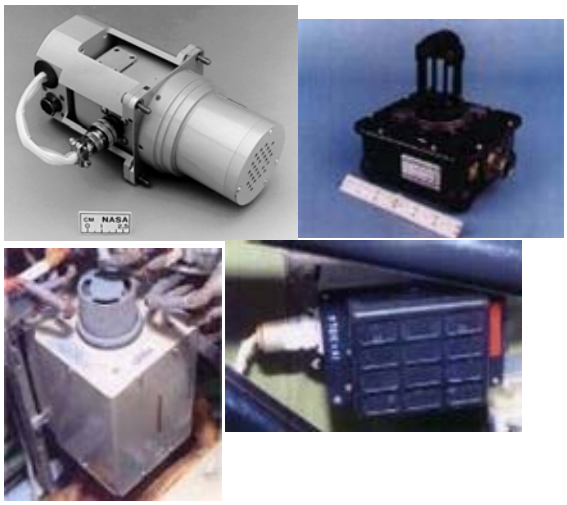


Figure 1: Current Spacecraft smoke detectors. *Clockwise from top left: ionization smoke detector used in the NASA Orbiter fleet; light scattering detector used in the US lab on the ISS; ionization detector used in the Russian ISS Service Module; and ionization detector used in the Russian ISS FGB Module.*

In the Mercury, Gemini and Apollo missions, the crew quarters were limited and mission durations were short, consequently it was considered reasonable that the astronauts would rapidly detect any fire. The Skylab module, however, included approximately 30 UV-sensing fire detectors [3]. The US Space Shuttle and the Russian segments of the ISS all use ionization smoke detectors while the US, JEM and Columbus labs of the ISS use light scattering detectors. The systems currently in use are shown in Figure 1.

Light scattering devices are most sensitive to particulate whose size is of the same order as the wavelength of the light used by the detector, typically particle diameters greater than 0.3 μm. Ionization devices are more sensitive to smaller particles. As reviewed by Urban et al. [2] this difference in size sensitivity is significant for some types of fires, consequently ionization and scattering detectors have different performance for normal-gravity fires. Proper implementation of smoke detection requires understanding of the smoke particle size distribution. Low-gravity smoke particle size data were not available when the existing systems were designed, consequently the designers used 1-g smoke detector designs. These design decisions led to detectors on the ISS and the Space Shuttle Orbiter that, in general, have non-overlapping particle size sensitivities.

As described by Friedman [4] there have been six overheated electrical component events in the NASA Orbiter fleet. None of these events spread into a real fire but as mission durations increase, the likelihood of failures increases. The experience on Mir in 1997 has shown that failure of oxygen generation systems can have significant consequences. As a result, improved understanding of spacecraft fire detection is critically needed. [5]

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Any effort to detect fires must be able to discriminate against the ambient background and nuisance signals. In the case of smoke detection, the background dust aerosol conditions must be considered. The only spacecraft background aerosol particulate measurements to date were made on the Orbiter on STS-32 in 1990 [6] and on the ISS. [2] The STS measurements reported a bimodal particle size distribution with ~ 40% of the particles in each of the 2.5 to 10 μm and >100 μm ranges. The other two ranges <2.5 μm and 10 to 100 μm were very lightly populated. These results showed substantially higher concentrations than typical indoor measurements (over a factor of 5 for the two most populated size ranges). The ISS results of Urban et al. [2] were a secondary result of an experiment to validate the operation of an aerosol particle counter in low-gravity. The measurements were made in the US Lab on Increment 10 and the particle count levels were at the noise level for the instrument (less than 5 particles/cc). The low particle levels in the < 2.5 μm bin for the STS and the low overall numbers for the ISS suggest a zone of opportunity for spacecraft fire detection since typical normal gravity fires produce substantial particulate in this size range[2,7]. However more complete particle size statistics are needed.

SAME EXPERIMENT CONCEPT

To date the only data on low-gravity smoke particulate were collected by the CSD experiment[2]. The CSD data set was not particularly large as the size statistics were obtained from TEM data, critical details were not obtained for the liquid droplet aerosols and the smoke flow conditions were not well characterized. To rectify these concerns, another experiment, the Smoke Aerosol Measurement Experiment (SAME) was developed. The SAME experiment sought to avoid the problems experienced by the CSD experiment by obtaining the particulate size statistics on-orbit with a reduced dependence upon sample return to Earth. This is a challenging endeavor because existing aerosol instrumentation is typically large, incompatible with spacecraft experiment constraints, and may require substantial sample return to Earth. As will be described below, an alternative approach was employed that used three discrete instruments to measure separate moments of the size distribution. When combined, these

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where dN is the number of particles per cm^3 with diameter between D and D + dD. The moments of interest consist of the number concentration, M_0, the first moment M_1, and the volume or mass concentration moment, M_3 As defined in Eq. 2.

$$M_i = \int D^i f_N(D) dD \quad i = 0,1,3 \quad (2)$$

The first moment which can also be thought of as the "diameter concentration" or integrated diameter-per-unit-volume is approximately proportional to the signal from an ionization smoke particulate detector. For particles in the Mie scattering regime, particles sizes from 0.3 λ to about 3 λ (~0.2 to 2.0 μm for a red laser) the light scattering signal is approximately proportional to the third moment. From these moments, and a measurement of M_0 from a condensation nuclei counter, two mean diameters can be computed: the count (arithmetic) mean diameter $D_{0.5}$ or $\bar{d}$, which is equal to M_1/M_0 and the diameter of average mass $D_{1.5}$ or $d_{\bar{m}}$, which is equal to $(M_3/M_0)^{1/3}$. The log-normal size distribution is widely used for describing the size distribution of aerosols including non-flaming smoke. For smokes, as in most aerosols, the bulk of the number concentration is associated with smaller particles.[9, 10] The form of this distribution function is the same as the normal distribution except that the diameter is replaced with the ln D so that one has

$$f_N(D) = \frac{N_t}{(2\pi)^{1/2} D \ln \sigma_g} \exp\left(- \frac{(\ln D - \ln D_g)^2}{2 \ln^2 \sigma_g} \right) \quad (3)$$

where N_t is the total number concentration of the aerosol (=M_0), and D_g and σ_g are the geometric mean number diameter and geometric standard deviation defined by

$$\ln D_g = \int_0^\infty \ln D f_N(D) dD / \int_0^\infty f_N dD \quad (4)$$

$$\ln \sigma_g = \left[\int_0^\infty (\ln D - \ln D_g)^2 f_N(D) dD / \int_0^\infty f_N(D) dD \right]^{1/2} \quad (5)$$

For the log-normal distribution, one finds that the various diameter definitions are related to the geometric mean number diameter, D_g, via the equation: [9, 10]

$$D_p = D_g \exp(p \ln^2 \sigma_g) \quad (6)$$

For the count mean diameter, $D_{0.5}$, and the diameter of average mass, $D_{1.5}$, the corresponding values of p are 0.5 and 1.5. As an example, for a log-normal distribution with D_g=1.0 μm and σ_g=2.0, the corresponding values of $D_{0.5}$ and $D_{1.5}$ are 1.27 μm and 2.06 μm, respectively. Using equation (6), one can relate σ_g to the ratio of $D_{1.5}$ and $D_{0.5}$ via the equation:

$$\sigma_g = \exp\left(\ln(D_{1.5} / D_{0.5})\right)^{1/2} \quad (7)$$

In summary, by combining these three measured moments of the size distribution, M_0, M_1, and M_3, it is possible to compute three mean diameters of the size distribution, $D_{0.5}$, $D_{1.5}$, and D_g, and the geometric standard deviation, σ_g. Validation of this approach is discussed in Cleary, Weinert and Mulholland. [8] These statistics provide a strong basis for design of spacecraft smoke detectors.

SAME EXPERIMENT DESIGN

The experiment was developed as a Microgravity Science Glovebox (MSG) payload under the ISS research program. A schematic of the experiment hardware appears in Fig. 2 and an assembled photograph in Fig. 3. Smoke was generated by overheating a small sample of material in the smoke generation duct for approximately 60 seconds. During this interval, controlled flow over the sample was

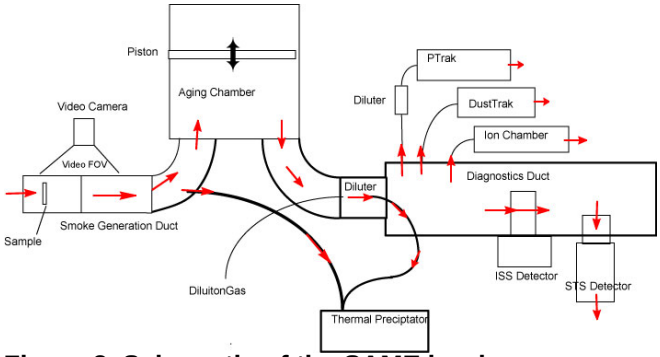


Figure 2. Schematic of the SAME hardware.



Figure 3: Engineering model of the SAME hardware installed in a MSG simulator. *The sample carousel is on the left side, the large vertical cylinder is the aging chamber and the moment instruments are on the right side. (Clockwise from top left): Dynamic Diluter for the zeroth moment instrument, Commercial Diagnostics Unit A (first and third moments), Commercial Diagnostics Unit B, Zeroth Moment instrument. The ISS and STS smoke detectors are visible to the left of the aging chamber.*

induced by a moving piston in the aging chamber which drew the smoke into the chamber. The smoke was held in the chamber for a predetermined time, allowing the particles to coagulate. The smoke was then pushed by the piston into the diagnostics duct where the moment instruments made their measurements. Also installed in the diagnostics duct were Space Shuttle and ISS smoke detectors. As the smoke was monitored by the moment instruments, a sample of the smoke particles was deposited on Transmission Electron Microscope (TEM) grid via a thermal precipitator which uses thermophoresis to deposit the particles on the grids.

Smoke was generated by resistively heating small samples of spacecraft materials. Sample heating was achieved by heating a steel wire that was wrapped around an approximately 3mm diameter cylinder of the sample material. The steel wire was resistively heated and the wire resistance was monitored to control its temperature. This technique was found to produce repeatable weight loss from sample to sample. Groups of six preweighed samples were installed in carousels (Fig. 4) that the crew installed in the SAME hardware. For each test, a new sample was moved into place and the sample heating and the flow were synchronized to provide a steady air stream over the sample during the heating interval. The system was designed to provide air flow rates ranging from 2 to 8 cm/s. To emulate a variety of fire scenarios, tests were conducted at different sample temperatures, air flow rates, and smoke aging durations. Five material were tested, Teflon™, Kapton™, silicone rubber, lamp wick (cellulose) and a reference material (dibutyl phthalate).



Figure 4. Photograph of a sample carousel. *Carousel contains an assortment of Teflon, Kapton, lamp wick, and silicone rubber. The samples were supported and heated by the same stainless steel wire coil.*

INSTRUMENTS

The moment measurements were made using an assembly of three separate instruments. Two are industrial hygiene instruments manufactured by TSI Inc. and one is a modified residential smoke detector. The zeroth-moment instrument is a condensation nuclei counter P-Trak™ (TSI Inc.). This device operates by passing the aerosol-laden particle stream through a volume saturated with isopropanol vapor and then into a cooler region where the vapor condenses onto the particles raising them all to a size where they are readily counted by a light scattering device. This instrument is very robust and operates over a range of 0 to 10^5 particles/cm^3 and 20 nm to 5 µm diameter. Some dilution is required since the smoke concentration ranges from about 0.5×10^6 to 5×10^6. There was also a concern that the isopropanol condensate would not return to the wick in low-gravity. [2] To mitigate this issue, the condensing section of the device was modified with very small grooves to improve conductance of the condensate back to the wick. These changes were tested in a separate space experiment with good results indicating the modified device could be used successfully in low gravity [2]. The device was repackaged into an enclosure (Fig. 3) and the internal air pump was replaced with a choked-flow orifice that vented into the MSG vacuum resource.

The first and third moment instruments were packaged together in a single enclosure and two of these combined enclosures were used simultaneously during SAME operations. The first-moment instrument is the ionization chamber from a residential smoke detector which uses an alpha-particle emitter to generate ions in a region inside a DC electric field. The sensor measures the voltage of a reference electrode inside this field. The presence of particulate changes the charge distribution (and thereby the reference electrode voltage) as a result of the attachment of the ions to the particulate. The

required particle concentrations are on the order of 10^5 particles / cm^3 and no sample dilution was required. The sensor was enclosed in a chamber through which the smoke was drawn at a controlled velocity by the MSG vacuum resource.

The third-moment instrument is a light scattering device DustTrak™ (TSI Inc.). The device uses a 90 degree light scattering signal to quantify the aerosol mass density. For terrestrial dust particulate this signal correlates well with the mass concentration, however additional compensation will be needed to account for the range of particle sizes that will be seen in the SAME experiment. The device operating range is from 0.001 to 100 mg /m^3. These devices are equipped with an aerodynamic impactor at the inlet which captures particles larger than the selected size. The SAME experiment included 2 DustTraks™ in separate enclosures. This made it possible to use one with a 1 µm impactor and one with a 10 µm impactor. The difference in the signal from these two devices provides a measure of the fraction of the particulate that was larger than 1 µm. In some cases dilution was required owing to the high smoke concentration levels.

As the aging chamber is filled and as the aerosol is pushed out to the moment instruments, a small aerosol stream is drawn off to the thermal precipitator, (Fig. 5). Inside the precipitator, the aerosol passes between a Transmission Electron Microscope (TEM) grid and a heated wire. Thermophoretic forces move particles to the grid surface where they adhere. After the mission the grids are removed from the assembly and examined in a TEM to obtain an independent determination of the particle size and morphology.

The smoke in the diagnostics duct is also sampled by an ISS smoke detector and a Space Shuttle smoke detector at the same time the moment instruments are making their measurements. These instruments are visible in Figure 3.

EXPERIMENTAL RESULTS

The system was installed in the Microgravity Science Glovebox, an ISS facility that provides many resources including: containment, power, data, video and uplink commanding. The hardware is shown installed in the MSG in Fig 6. Overall 30 sample materials were tested. These were comprised of six samples each of 5 materials: Teflon™, Kapton™, silicone rubber, cellulose, and dibutyl-phthalate deposited on a porous wick. These samples were tested at different airflow rates, pyrolysis temperatures, and smoke aging durations. The analog signals from the diagnostics were digitized at 1 Hz. Figure 7 contains the signal traces from a typical run for silicone rubber. The difference in the signals for the two DustTrak's indicates that a substantial portion of the aerosol mass is in particles that are larger than 1 µm. For comparison, Fig. 8 contains similar data for a Teflon run where little difference is evident between the two

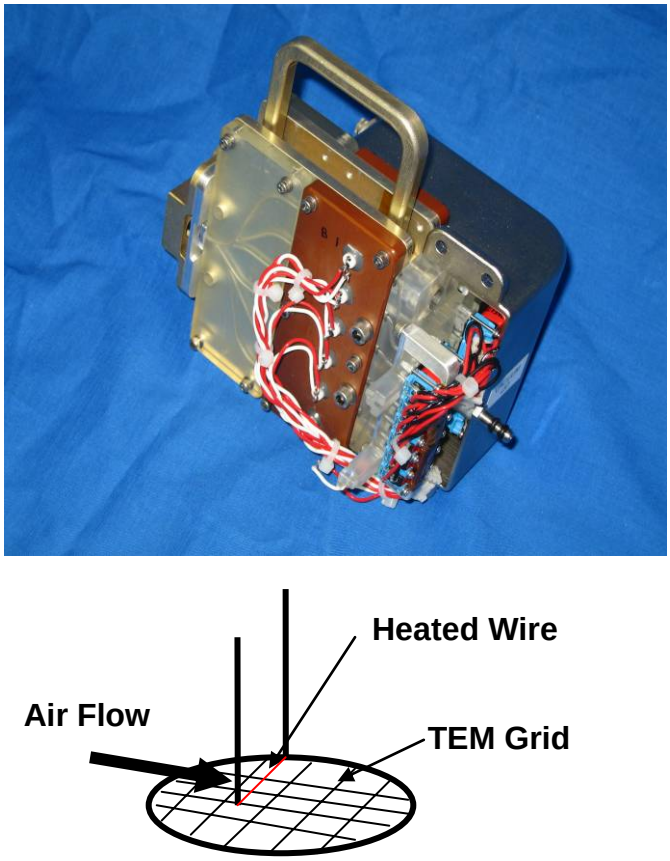


Figure 5: Thermal Precipitator Module. *Top image is a photograph of a partially disassembled Thermal Precipitator Module with the inlet manifold exposed on the left, the hot wire leads in the red Vespel block in the middle and the outlet valves on the right. The drawing below displays the position of the hot wire above the TEM grid with the air flow passing over both. The heated wire drives aerosol particles onto the grid via thermophoresis.*

DustTrak instruments, indicating that the bulk of the aerosol mass is in particles below 1 µm. The effect of aging is shown in Fig. 9 where, consistent with particle coagulation with limited wall loss, the overall number count (Ptrak) decreases substantially while the mass concentrations (DustTraks) remain relatively steady.

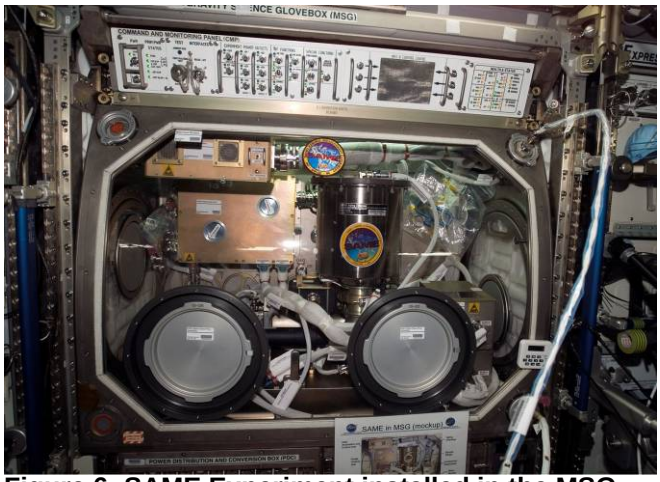


Figure 6: SAME Experiment installed in the MSG Facility.

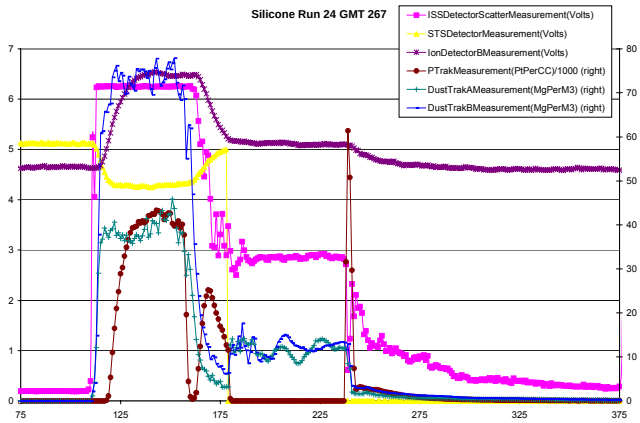


Figure 7: Instrument traces for a typical silicone rubber run. *Base levels are at the beginning and end of the trace. The actual sampling interval is between 125 and 160 seconds. The differences in the rise times of each instrument are due to the smoke transit time to the instrument. The large difference between DustTrak A and DustTrak B is indicative of the substantial portion of the particle distribution that is larger than 1μm (the cutoff for the impactor on DustTrak A).*

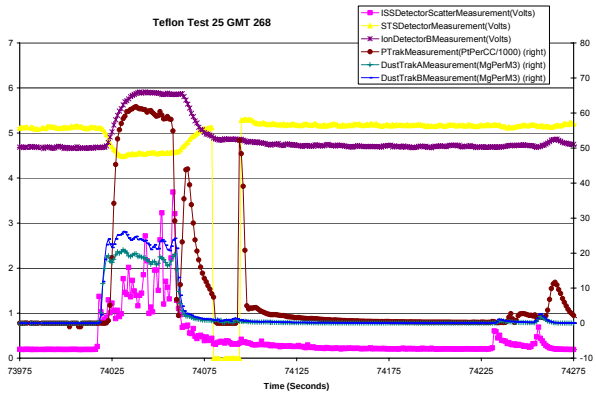


Figure 8: Instrument traces for a typical Teflon run. *Base levels are at the beginning and end of the trace. The actual sampling interval is between 74025 and 74060 seconds. The small difference between DustTrak A and DustTrak B is indicative that the most of the particle distribution is smaller than 1μm (the cutoff for the impactor on Dust Trak A).*

It is also interesting to compare the response of the ISS and STS detectors for silicone rubber and Teflon. In figure 7, both the ISS detector and the STS detectors show strong signals. (Note that the presence of smoke yields a positive deflection of the ISS signal and a negative deflection on the STS signal.) This is reasonable given the broader size distribution for silicone smoke containing significant numbers of both large and small particles. For the Teflon smoke comprised primarily of particles having diameters less than 1 micron as shown in figure 9, the STS detector provided a strong signal. Conversely, the signal from the ISS detector was more erratic. This will be re-visited after considering the preliminary data from the moment instruments.

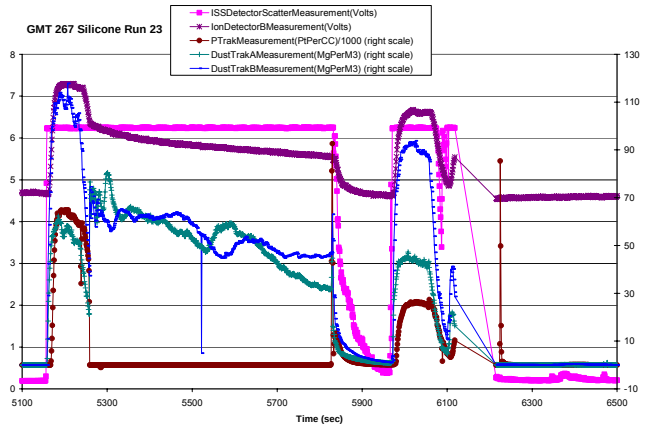


Figure 9: Instrument traces for a typical smoke aging run for silicone rubber. *Base levels are at the beginning and end of the trace. The first sampling interval is between 5200 and 5250 seconds and the post aging sampling interval is between 6000 and 6050 seconds. Consistent with particle coagulation with limited wall loss, the overall number count (Ptrak) decreases substantially while the mass concentrations (DustTraks) remain relatively steady.*

At this time the samples and TEM grids have been received back from the ISS; TEM and mass loss analyses have been initiated but not completed. Validation of the instrument calibration is underway and ground-based comparison (1-g) runs are being conducted. The zeroth and first moment instruments have a limited dependence on particle diameter so they may be directly calibrated. The third moment instrument (the DustTrak™) has a significant dependence on both the particle diameter and refractive index. An algorithm has been implemented to use the results of a Mie scattering calculation with the measured results to iteratively converge on the value of the third moment that best fits the DustTrak™ signal and the other two moment measurements. This third moment value will be used to compute the geometric mean diameters and standard deviations.

Average values of the count mean diameter for all of the unaged results for each material are tabulated in Table 1 along with the ratio of the signal from the two DustTraks. Recall that the impactor used in DustTrak A removed particles greater than approximately 1 micron; the impactor for DustTrak B removed particles greater than about 10 microns (although few particles greater than 10 microns should be present in these conditions). Although the count mean diameters are all in the 100 to 200 nm range, interpreting particle sizes by only one statistic can be deceptive due to the nature of the particle size distribution. In general, since the mass increases with the cube of the diameter, the larger particles do not affect the count mean diameter (CMD) as much as they affect the light scattering signal which scales with the particle mass. Consequently, although the CMD for silicone rubber is 227 nm, almost half the particle mass is larger than 1000 nm, as evidenced by the ratio of the DustTrak instruments. The lamp wick showed similar behavior.

Material	Count Mean Diameter (nm)	Ratio of DustTrak B/DustTrak A
Silicone Rubber	227	1.92
Teflon™	158	1.22
Lamp wick	172	1.79
Kapton™	128	1.26

Table 1: Average Count Mean Diameters and ratios of the Dust Trak instruments. *Dust Trak A had a 1 µm cutoff impactor while the cutoff for DustTrak B was 10 µm.*

On the other hand, the Kapton and Teflon showed substantially narrower particle size distributions with little mass in particles larger than 1000 nm. These low populations of larger particles are consistent with the less effective detection by light scattering devices which was seen for some of the Teflon tests and with the signals obtained from the STS and ISS detectors. All of the tests conducted in this series of experiments showed strong signals on the Space Shuttle detector (ionization) despite the larger particle size distributions seen for the lamp wick(cellulose) and silicone rubber. The CSD experiment [2] reported poor detection by the Space Shuttle detector for nonflaming overheat of cellulose and silicone rubber (indicating larger particle sizes). The reduced signal on the Space Shuttle detector for CSD suggests that under some reasonable pre-fire conditions the smoke size distribution can extend to diameters beyond those seen in the SAME experiment. The detection of smoke across the size range seen in SAME and CSD, while still discriminating against dust (which tends to be large) will probably require a detection system that can discriminate the particle size distribution to at least a limited degree. Two effective means of achieving this size discrimination are to employ a detector that measures more than one moment of the size distribution or to use a detector that can classify the particles into size ranges.

CONCLUSIONS

The SAME experiment operated successfully on the ISS during Increment 15. The preliminary results from the experiments indicate that substantial differences in the size distributions of the smoke particles are evident for the four materials that were tested. For the conditions of the SAME experiment, all samples produced significant numbers of sub-micron particulate that are better detected using an ionization smoke detector. Considering the results from both the SAME and CSD experiments, the only two fire detection experiments conducted in low-gravity, broader smoke particulate size distributions can be produced from credible pre-fire overheat events, depending on the conditions. These results suggest that detection methods that can measure more than one moment of the size distribution may show more successful detection and alarm rejection than single moment detectors.

ACKNOWLEDGMENTS

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Certain commercial entities, equipment, or materials may be identified in this document in order to describe an experimental procedure or concept adequately. Such identification is not intended to imply recommendation or endorsement by the National Aeronautics and Space Administration or the National Institute of Standards and Technology, nor is it intended to imply that the entities, materials, or equipment are necessarily the best available for the purpose.

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ACRONYNMS

CMD: Count Mean Diameter

CSD: Comparative Soot Diagnostics

DAFT: Dust Aerosol measurement Feasibility Test

ISS: International Space Station

MSG: Microgravity Science Glovebox

SAME: Smoke Aerosol Measurement Experiment

STS: Space Transportation System

TEM: Transmission Electron Microscope

Spacecraft Fire Detection: Smoke Properties and Transport in Low-Gravity

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National Center for Space Exploration Research, Cleveland, Ohio 44135

Results from a recent smoke particle size measurement experiment conducted on the International Space Station (ISS) are presented along with the results from a model of the transport of smoke in the ISS. The experimental results show that, for the materials tested, a substantial portion of the smoke particles are below 500 nm in diameter. The smoke transport model demonstrated that mixing dominates the smoke transport and that consequently detection times are longer than in normal gravity.

I. Introduction

The absence of buoyant flow increases the residence time in microgravity fires causing substantial changes in flame structure and consequently changes the properties of the produced smoke. In particular, the increased residence time in the high temperature zone where the smoke is formed, can increase the opportunity for growth in the size of the smoke particulate. However, no measurements of the properties of smoke from microgravity fires had been made at the time when the detectors were designed for both the space shuttle and the ISS. This absence of data caused the designers to base the design on 1-g information and the current state of the art. As a result, the two detectors were designed using different detection technologies which have very different particle size sensitivities. The objective of this work was to make sufficient measurements of smoke from spacecraft fires to enable improved design of future detectors.

A. Smoke Background

Smoke is a general term that encompasses aerosol materials produced by a number of processes. In particular it can include unburned, recondensed, original polymer or pyrolysis products that can be either liquid or solid, hydrocarbon soot, condensed water vapor, and ash particles. Ash and soot particles dominate the smoke particulate in established flaming fires while unburned pyrolysis products and recondensed polymer fragments are produced by smoldering and pyrolysis in the early stage of fire growth. Given the constrained space on any spacecraft, the target

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[6] Visiting Professor of Mechanical Engineering, Glenn L. Martin Hall.
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for the fire detection system is necessarily the early phase and not established flaming fires; consequently, the primary target for detection is the pyrolysis products and not the soot and ash.

B. Prior Spacecraft Systems

In the Mercury, Gemini and Apollo missions, the crew quarters were limited and mission durations were short, consequently it was considered reasonable that the astronauts would rapidly detect any fire. The Skylab module, however, included approximately 30 UV-sensing fire detectors.[1] These devices were limited to line-of-sight and were reported to have difficulties with false alarms.

C. Space Shuttle Detectors

At the time the Space Shuttle Orbiter (Space Transportation System (STS)) was developed, smoke detection for terrestrial applications was becoming commonplace.[2] In particular, ionization detectors were becoming readily available but photoelectric (scattering or obscuration detectors) were generally unavailable due to the difficulty of producing stable light sources. There was no data available concerning the smoke or dust particle size distribution in low-gravity and the database of normal-gravity smoke characteristics was a fraction of what is available today. Consequently, the STS smoke detector (Fig. 1) employed a variant of the accepted ground based approach (ionization detectors). The Orbiter has nine particle-ionization smoke detectors in avionics cooling air return lines in the mid-deck and flight deck and SpaceLab had six additional particle-ionization smoke detectors in the avionics lines.[3] While the design rationale is not completely known, Celesco™ (later Brunswick Defense™) based their design on data that suggested incipient fires could be discriminated by looking for particulate in the 0.4 to 0.7 μm range.[4] Use of a pump provided the opportunity to employ a particle separator allowing rejection of particulate larger than 1μm. This air pump increased the power requirements and reduced the operational life for the detectors.

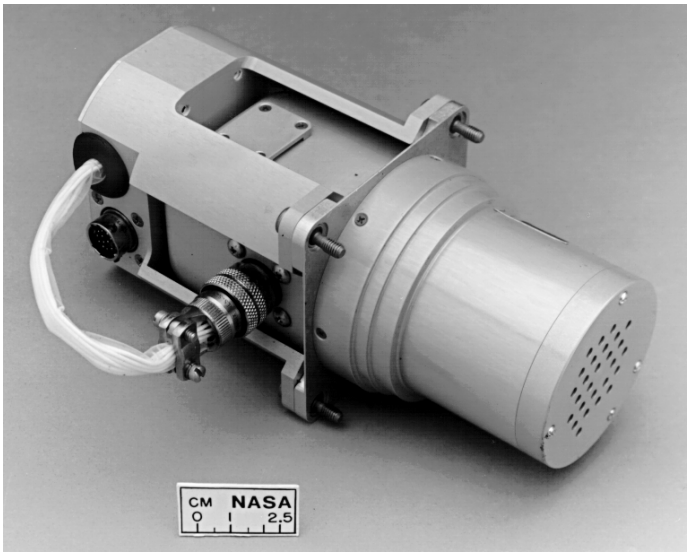


Figure 1. Brunswick Defense™ smoke detector used in the NASA Orbiter fleet. The inlet is on the right and the gas is expelled out the small plate on the top.

D. ISS Detectors

The International Space Station (ISS) smoke detector design took advantage of the availability of stable laser-diode light sources, mirroring a trend in terrestrial detectors toward light scattering devices. This provided the opportunity to produce a detector that required substantially less power, (1.5 Watts versus 9 Watts for the Orbiter detector)[5] and, with no moving parts, was quieter with a much longer operational life. The choice was supported by data indicating that early smoldering fires produce larger particulate than established flaming fires.[2] The designated detectors (Fig. 2) for the ISS are laser-diode, forward-scattering, smoke or particulate detectors. Their sensitivity is greatest for particles larger than 1 μm with sensitivity extending down to 0.6 μm. The current requirements for the ISS call for two detectors in the open area of the US Lab module, and detectors in racks that have cooling air-flow.[6]

The design developed for the ISS by Allied Signal™ (Fig. 2) consists of a 2-pass laser diode to sense forward scattered light (30 degrees). There is also a zero degree obscuration system that is used as a measure of the beam strength. The system is designed to alarm based on the magnitude of the scattered light signal referenced to the beam intensity. The system is less sensitive to particles smaller than the wavelength of the laser (near IR) than it is to larger particles. The minimum reported sensitivity is 0.3 μm.[5] On the other hand, for a fixed mass concentration of smoke aerosol; the STS ionization smoke detector is more sensitive to smaller particle sizes.

As a result of these design decisions, the ISS and the Orbiter detectors are sensitive to particles in different size ranges. In both cases the designs were based upon the best available data however, due to the complete absence of low-g data concerning the nature of particulate and radiant emission from incipient and fully developed low-g fires, different conclusions were drawn concerning the optimal design for spacecraft fire detection.

As described by Friedman[7] there have been six overheat and failed component failures in the NASA Orbiter fleet. None of these events spread into a real fire but as mission durations increase, the likelihood of failures increases. The experience on Mir in 1997 has shown that failure of oxygen generation systems can have significant consequences. As a result, improved understanding of spacecraft fire detection is critically needed.[8]

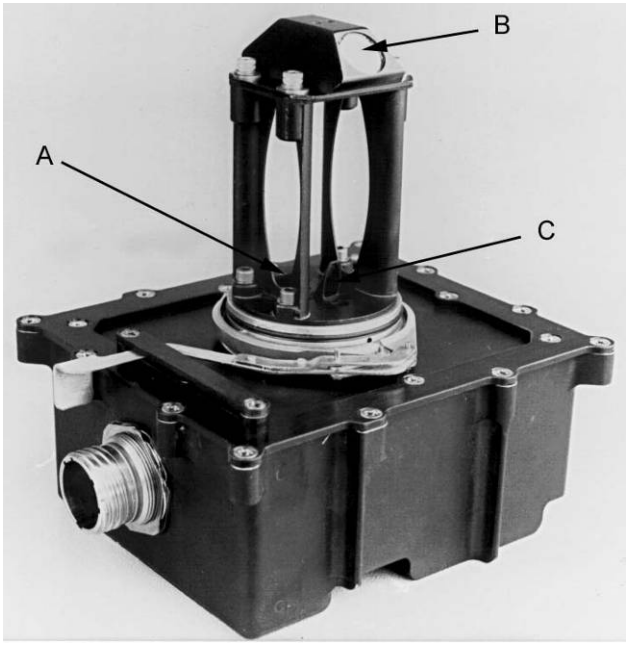


Figure 2. Allied Signal/Honeywell light-scattering smoke detector used in the ISS. The near IR laser beam emerges from the enclosure into the top assembly (A) and is reflected by two mirrors (one visible at top right (B)) and is then reflected back to the sensors in the enclosure (C). One sensor detects the forward scattered light and is referenced by another sensor that looks directly at the incident beam.

E. Background in Low Gravity Smoke

The salient result on smoke particle size determination in low-gravity reported by Urban et al.[9] is that particulate produced by low-gravity flames (soot or unburned fuel particles) tends to have larger size particles than in normal gravity. Results from the CSD (Comparative Soot Diagnostics) Experiment[9] which studied smoke properties in low-gravity from several spacecraft materials suggested that liquid smoke particles could achieve sizes larger than 1 μm while solid particulate remained in the sub-micrometer range. The combined impact of these limited results and theoretical predictions is that, as opposed to extrapolation from 1-g data, direct knowledge of low-g combustion particulate is needed for more confident design of smoke detectors for spacecraft.

F. Spacecraft Atmosphere Dust Background

Any effort to detect fires must be able to discriminate against the ambient background and nuisance signals. In the case of smoke detection, the background dust aerosol conditions must be considered. The only spacecraft background aerosol particulate measurements to date were made on the Orbiter on STS 32 in 1990[10] and on the ISS.[9] The STS measurements included the results from two cascade impactors and a light scattering device. The two impactors reported a bimodal particle size distribution with ~ 40% of the particles in each of the 2.5 to10 μm and >100 μm ranges. The other two ranges < 2.5 μm and 10 to 100 μm were very lightly populated. Each impactor sampled approximately 15 m^3 of air over approximately 30 hours. These results showed substantially higher concentrations than typical indoor measurements (over a factor of 5 for the two most populated size ranges). These data were supported by the light scattering instrument which made 17 measurements in 12 locations, all of these reported mass concentrations ranging from 50 to 70 $\mu g/m^3$. The ISS results of Urban et al.[9] were a secondary result of an experiment to validate the operation of an aerosol particle counter in low-gravity. The measurements were made in the US Lab on Increment 10 and the particle count levels were at the noise level for the instrument (less than 5 particles/cc). The low particle levels in the < 2.5 μm bin for the STS and the low overall numbers for the ISS suggest a zone of opportunity for spacecraft fire detection since typical normal gravity fires produce substantial particulate in this size range.[2,11] However more complete particle size statistics are needed.

II. SAME Experiment

Although the CSD experiment produced useful information concerning the likely size of spacecraft smoke particulate, the data set was not particularly large as the size statistics were obtained from TEM data and critical details were not obtained for the liquid droplet aerosols. To rectify this concern, another experiment (Smoke Aerosol Measurement Experiment: SAME) was developed. The SAME experiment sought to avoid the problems experienced by the CSD experiment by obtaining the particulate size statistics on-orbit with a reduced dependence upon sample return to Earth. This is a challenging endeavor because existing aerosol instrumentation is typically large, incompatible with spacecraft experiment constraints, and may require substantial sample return to Earth. As will be described below, an alternative approach was employed that used three discrete instruments to measure separate moments of the size distribution. When combined, these moments provide useful aggregate statistics of the size distribution. The measurements were made using smoke generated by overheated spacecraft materials in much the same manner as the CSD experiment however the sample temperature, flow field, and particle aging time were more rigorously controlled.

A. Moment Method

The approach used by the SAME experiment is termed the 'moment method' for convenience.[12] As will be described below, the approach consists of measuring three moments of the size distribution (zeroth, first and third) and using the properties of the log-normal distribution to estimate the geometric mean diameter and the standard deviation.

The average particle size and an estimate of the width of the size distribution will be estimated from various moments of the size distribution. The number distribution, $f_N(D)$, is defined as

$$f_N(D) = \frac{dN}{dD} \tag{1}$$

where dN is the number of particles per cm^3 with diameter between D and D + dD. The moments of interest consist of the number concentration, M_0, the first moment M_1, and the volume or mass concentration moment, M_3.

$$M_i = \int D^i f_N(D) dD \quad i = 0,1,3 \tag{2}$$

The first moment which can also be thought of as the "diameter concentration" or integrated diameter per unit volume is approximately proportional to the ionization detector moment (signal). For particles in the Mie scattering regime, particles sizes from 0.3λ to about 3λ (~0.2 to 2.0 μm for a red laser) the light scattering signal is approximately proportional to the third moment. From these moments, and a measurement of M_0 using a condensation nucleus counter, two mean diameters can be computed: the count (arithmetic) mean diameter $D_{0.5}$ or $\overline{d}$, which is equal to M_1/M_0 and the diameter of average mass $D_{1.5}$ or $\overline{d}_m$, which is equal to $(M_3/M_0)^{1/3}$. The log-normal size distribution is widely used for describing the size distribution of aerosols including non-flaming smoke. As in most aerosols; the bulk of the number concentration is associated with smaller particles.[13, 14] The form of the distribution is the same as the normal distribution except that the diameter is replaced with the ln D so that one has

$$f_N(D) = \frac{N_t}{(2\pi)^{1/2} D \ln \sigma_g} \exp\left(-\frac{\left(\ln D - \ln D_g\right)^2}{2 \ln^2 \sigma_g} \right) \tag{3}$$

where N_t is the total number concentration of the aerosol (=M_0), and D_g and σ_g are the geometric mean number diameter and geometric standard deviation defined by

$$\ln D_g = \int_0^\infty \ln D f_N(D) dD \Big/ \int_0^\infty f_N dD \tag{4}$$

$$\ln \sigma_g = \left[\int_0^\infty (\ln D - \ln D_g)^2 \, f_N(D) dD \Big/ \int_0^\infty f_N(D) dD \right]^{1/2} \tag{5}$$

For the log-normal distribution, one finds that the various diameter definitions are related to the geometric mean number diameter, D_g, via the equation:[13, 14]

$$D_p = D_g \exp(p \ln^2 \sigma_g) \tag{6}$$

For the count mean diameter, $D_{0.5}$, and the diameter of average mass, $D_{1.5}$, the corresponding values of p are 0.5 and 1.5. As an example, for a log-normal distribution with D_g=1.0 μm and σ_g=2.0, the corresponding values of $D_{0.5}$ and $D_{1.5}$ are1.27 μm and 2.06 μm, respectively. Using equation (6), one can relate σ_g to the ratio of $D_{1.5}$ and $D_{0.5}$ via the equation:

$$\sigma_g = \exp\left(\ln(D_{1.5} / D_{0.5})\right)^{1/2} \tag{7}$$

By combining these three moments it is possible to compute three mean diameters of the size distribution and the geometric standard deviation. Validation of this approach is discussed in Cleary, Weinert and Mulholland.[3] These statistics provide a strong basis for design of spacecraft smoke detectors.

B. Instruments

These measurements were made using an assembly of three separate instruments. Two are industrial hygiene instruments manufactured by TSI and one is a modified residential smoke detector.

The zeroth-moment instrument is a condensation nuclei counter P-Trak™ (TSI Inc.). This device operates by passing the aerosol-laden particle stream through a region where it is saturated with isopropanol vapor and then into a cooler region where the vapor condenses onto the particles raising them all to a size where they are readily counted by a light scattering device. This instrument is very robust and operates over a range of 0 to 105 particles/cm^3 and 20 nm to 1 μm diameter. Some dilution is required since the smoke concentration ranges from about 0.5 × 10^6 to 5 × 10^6. There was also a concern that the isopropanol condensate would not return to the wick in low-gravity.[9] To mitigate this issue, the condensing section of the device was modified with very small grooves to improve conductance of the condensate back to the wick. These changes were tested in a separate space experiment with good results indicating the modified device could be used successfully in low gravity.[9]

The first-moment instrument is the ionization chamber from a residential smoke detector which uses an alpha-particle emitter to generate ions in a region inside a DC electric field which causes the ions to drift producing current. The presence of particulate reduces the current as a result of the attachment of the ions to the particulate. The mobility of the charged aerosol is too small for it to be collected on the ionization chamber electrode. The required particle concentrations are on the order of 105 particles / cm^3 and no sample dilution was required.

The third-moment instrument is a light scattering device DustTrak™ (TSI Inc.). The device uses a 90 degree light scattering signal to quantify the aerosol mass density. For terrestrial dust particulate this signal correlates well with the mass concentration, however additional compensation will be needed to account for the range of particle sizes that will be seen in the SAME experiment. The device operating range is from 0.001 to 100 mg /m^3. These devices are equipped with an aerodynamic impactor at the inlet which captures particles larger than the selected size. The SAME experiment included 2 DustTraks™ one with a 1 μm impactor and one with a 10 μm impactor. The difference in the signal from these two devices provided a measure of the fraction of the particulate that was larger than 1 μm. In some cases dilution was required owing to the high smoke concentration levels.

A schematic of the assembled hardware appears in Fig. 3. Smoke was generated by overheating a small sample of material in the smoke generation duct for approximately 60 seconds. During this interval, controlled flow was induced by a moving piston in the aging chamber which drew the smoke into the chamber. The smoke was held in the chamber for a predetermined time, allowing the particles to coagulate. The smoke was then pushed by the piston into the diagnostics duct where the moment instruments made their measurements. Also installed in the diagnostics duct were space shuttle and ISS smoke detectors. As the smoke was monitored by the moment instruments, a

sample of the smoke particles was deposited on Transmission Electron Microscope (TEM) grid via a thermal precipitator which uses thermophoresis to deposit the particles on the grids.

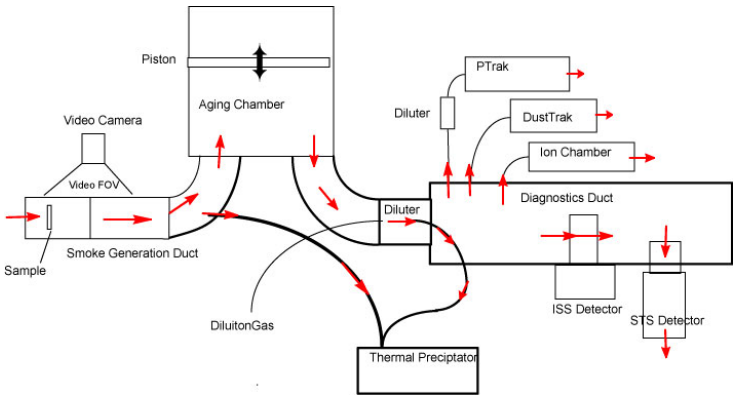


Figure 3. Schematic of the SAME hardware.

C. Experimental Results

The system was installed in the Microgravity Science Glovebox, an ISS facility that provides many resources including: containment, power, data, video and uplink commanding. The assembled hardware is shown in Fig 4. Overall 30 sample materials were tested. These were comprised of six samples each of 5 materials: Teflon™, Kapton™, silicone rubber, cellulose, and dibutyl-phthalate deposited on a porous wick. These samples were tested at different airflow rates, pyrolysis temperatures, and smoke aging durations. The analog signals from the diagnostics were digitized at 1 Hz. At this time the used samples and TEM grids have just been received back from the ISS so mass-balances and TEM analysis have not been initiated. The zeroth and first moment instruments have a limited dependence of particle diameter so they may be directly calibrated. The third moment instrument (the DustTrak™) has a significant dependence on the particle diameter and the refractive index. An algorithm has been implemented to use the results of a Mie Scattering calculation with the measured results to iteratively converge on the value of the third moment that best fits the DustTrak™ signal and the other two moment measurements. This work is currently underway so only results from the zeroth and first moment instruments will be presented here. As discussed above, the ratio of the first to the zeroth moment is the count mean diameter ($D_{0.5}$). Values of $D_{0.5}$ were calculated directly from the calibrated signals from the P-Trak and the Ion sensor and are shown in Table 1 for the baseline runs of the 4 solid materials with the percentage increase over the sizes from a comparable test in normal gravity.

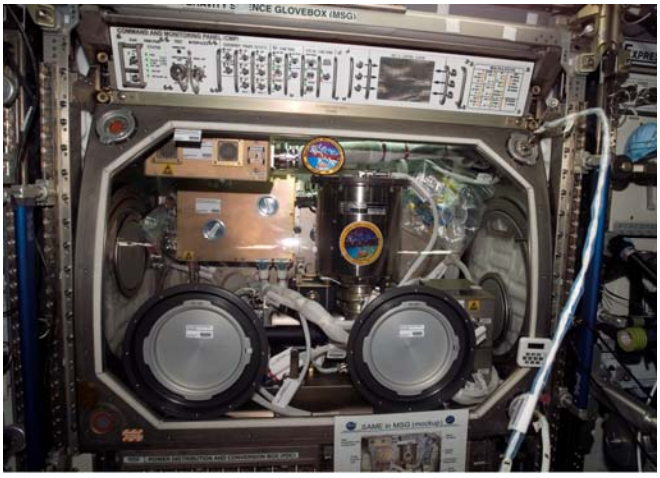


Figure 4. SAME Experiment installed in the MSG Facility

III. Smoke Transport Model

To assist in the interpretation of the SAME results and to provide a further understanding of the implementation of smoke detectors in spacecraft applications, a numerical model of the smoke transport and detection in the Destiny Laboratory (US Lab) of the ISS was developed. The details of the model are discussed by Brooker et al.[15] This study was conducted using FDS (Fire Dynamics Simulator) a fire modeling program developed by McGrattan et al.[16,17] at NIST. FDS (version 4.0.7 Linux) is a computational fluid dynamics (CFD) model that numerically solves a form of the Navier-Stokes equations for low speed, thermally driven flow with an emphasis on smoke and heat transport from fires.[16] A detailed description of the theoretical equations, underlying

Material	Count Mean Diameter (nm)	Count Mean Diameter (% larger than 1-g)
Silicone Rubber	263	53
Teflon™	223	
Lamp wick	248	65
Kapton™	114	58

Table 1: Count Mean Diameters for Baseline Runs with comparison to 1-g tests.

assumptions, and numerical solution can be found in the technical reference guide.[17] For this application, large eddy simulation (LES) was utilized in which the large scale eddies are computed directly and the sub-grid scale dissipative processes are modeled. Since the ISS is in low earth orbit, the gravitational acceleration was set to zero. As a result, the vorticity generated by buoyancy is no longer present in the momentum equation; therefore, the baroclinic torque was included to account for other sources of vorticity. A mixture fraction combustion model (a conserved quantity representing the fraction of material at a given point that originated in the fuel stream) was utilized for this analysis. All solid surfaces were designated as inert with a thermal boundary condition of 293 K. The number of grid cells spanning each coordinate direction was determined such that the cell dimension was approximately 5 cm in length for a total of 259,200 cells. The model layout is shown in Fig 5.

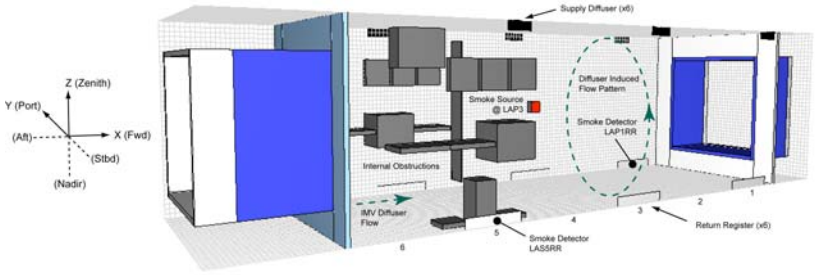


Figure 5. Computational Model Domain showing six overhead supply diffusers, six deck return registers, and one aft-port IMV diffuser for air ventilation. Induced flow patterns are indicated with dashed lines. Two area smoke detectors are located in front of LAP1 return register (LAP1RR) and LAS5 return register (LAS5RR). Also shown are internal obstructions representing on-orbit equipment and a smoke source at LAP3. ISPR sections are numbered one through six (fwd to aft) and labeled: port (P), starboard (S), nadir (N), or zenith (Z).

The smoke generation was modeled using a source positioned at the center of each rack location or ISPR (International Standard Payload Rack). The source was modeled as a propane burner with an area of 0.023 m^2 and heat flux of 57.6 kW/m^2 at three soot yield levels, producing smoke particulate outputs of 1.46, 2.95, and 8.47 mg/s. The alarm point for the ISS detector is on the order of 0.5 mg/m^3. The computational model was run at each smoke generation rate and rack location. The flow conditions in the lab were established to duplicate the operation of the ventilation system on orbit. The return and diffuser vanes are oriented to create an overall toroidal flow field that rotates counterclockwise (looking forward). Axial flow is induced near the floor on the port side by the Inter Module Ventilation (IMV) register on the aft bulkhead. For smoke generation locations at sections 5 and 6, the simulations were also run with and without the internal obstructions shown in Fig. 5. These internal obstructions were positioned to simulate typical installed hardware in the US Lab Module. The smoke source was activated 30 seconds after the simulation was initiated and the simulations were run for an additional 300 seconds. Figure 6 presents a reconstruction of the smoke distribution for an 8.5 mg/s generation rate case after 177 seconds. It is notable that other than the high concentration zone near the source, the smoke is fairly well mixed throughout the region without obstructions. The region with obstructions has lower smoke levels than the rest of the module. This scenario is substantially different from that seen in normal gravity where the smoke concentrates near the ceiling and mixing is generally not as complete as seen in Fig. 6.

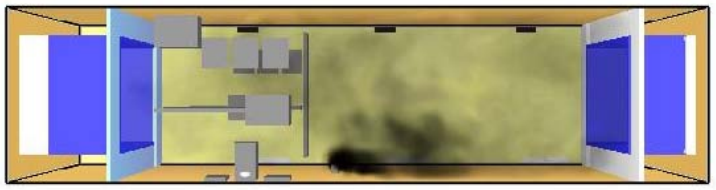
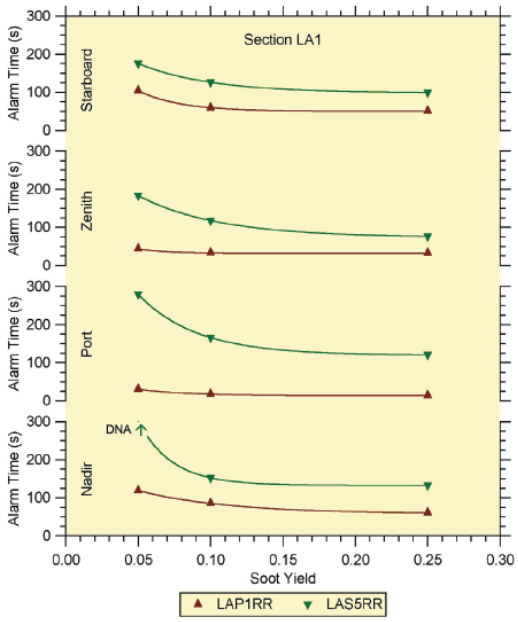


Figure 6: Representation of the smoke distribution 177 seconds after the smoke source was activated at location 4 on the floor. The smoke generation rate was 8.5 mg/s.

Figures 7, 8 and 9 show the alarm time response for sources at Section LA1 (Fig. 7), LA3 (Fig. 8) and LA6 (Fig.9). Evident in the figures is the variation in detection times with soot yield and source location. Although for LA3, for the zenith smoke source, the detectors alarmed at the same time, overall one detector typically alarmed significantly sooner than the other, particularly at low smoke yield levels. For LA1and LA3, the port detector was the first to alarm in most cases while by LA6 the starboard detector was the first to alarm. This behavior is consistent with the relative proximity of the detectors to the smoke sources. In Fig. 9, the impact of the obstructions is evident. For sources on the floor, the obstruction had no effect. This is consistent with expectation since the

floors are unobstructed and the flow is across the floor toward the detector. For cases on the starboard or zenith, the obstructions had a strong impact due to the longer transit time of the smoke to the detector.



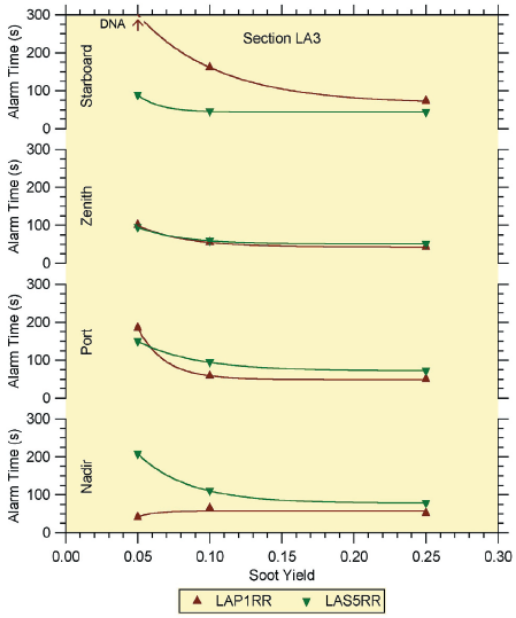


Figure 7 (left) and 8 (right): Smoke detector alarm time response to a variable strength smoke source. Data is from a circumferential profile at section LA1 (Fig. 7) and LA3 (Fig. 8). Note that a DNA (did not alarm) condition occurred for LA1 for the 0.05 soot yield on the N1 wall for LAS5RR and for LA3 0.05 soot yield on the S3 wall for LAP1RR.

IV. Conclusion

Smoke particulate produced in low-gravity was found to be typically 50% large in count mean diameter than similar conditions in normal gravity. The resultant sizes were all below 300 nm suggesting that discriminating smoke from spacecraft dust possibly could be achieved by detecting in the sub-micrometer range. Numerical modeling of smoke transport in the US Lab Module of the ISS predicted that actual detection times can be quite long and strongly dependent on location and the presence of obstructions.

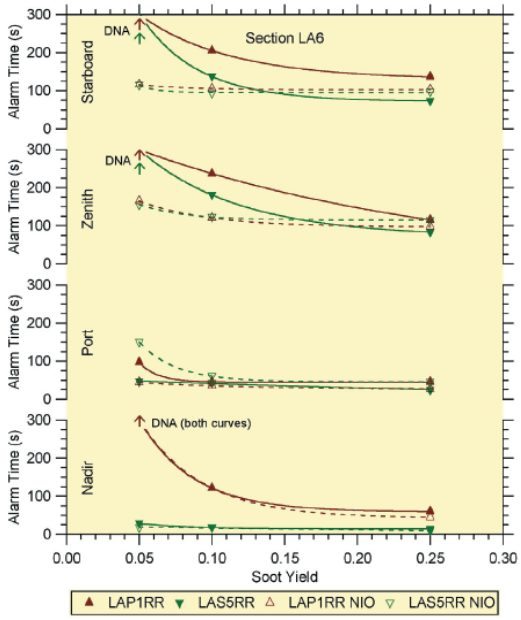


Figure 9. Smoke detector alarm time response to a variable strength smoke source. Data is from a circumferential profile at section LA6. Note that a DNA (did not alarm) condition occurred for the 0.05 soot yield on the S6, Z6, and N6 walls for LAP1RR, and S6 and Z6 walls for LAS5RR. Solid lines and symbols: with internal obstructions, dashed lines and open symbols: no obstructions.

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Certain commercial entities, equipment, or materials may be identified in this document in order to describe an experimental procedure or concept adequately. Such identification is not intended to imply recommendation or endorsement by the National Aeronautics and Space Administration or the National Institute of Standards and Technology, nor is it intended to imply that the entities, materials, or equipment are necessarily the best available for the purpose.

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